Control Theory

Chap 3: System Analysis

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Objectives:

1. Understand the importance and meanings of system stability;
2. Apply the Routh and Hurwitz criteria, the Bode diagram, the Polar plot, the Nyquist method, and the Lyapunov method to analyze the stability of open-loop and closed-loop system modelled by Laplace transformation equation;
3. Analyze the system stability modelled by differential equation, and the uncertain models.



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Chap 3: System Analysis

1. Introduction
2. Routh and Hurwitz Stability Criteria
3. Bode Diagram
4. Polar Plot
5. Nyquist Stability Criteria
6. Root Locus Method
7. Lyapunov Stability Criteria
8. Robust Stability
9. Chapter Conclusion

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1 Introduction

2 Routh and Hurwitz Stability Criteria

3 Bode Diagram

4 Polar Plot

5 Nyquist Stability Criteria

6 Root Locus Method

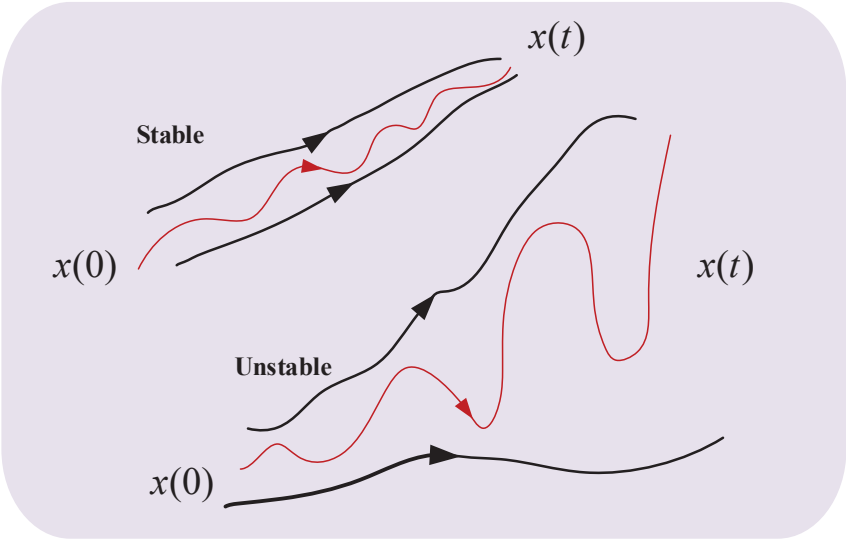
7 Lyapunov Stability Criteria

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9 Chapter Conclusion



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Stable v.s. Unstable

If a small perturbation arises in the system inputs or initial conditions,

1. a stable system will present small or limited modifications in its perturbed response;
2. an unstable system will make states or outputs grow unbounded or until the system disintegrates or saturates.

Definition (Bounded-Input Bounded-Output Stability (BIBO))

For a system

$$\frac{Y(s)}{U(s)} = G(s) \tag{1}$$

The bounded-input bounded-output (BIBO) stability implies that for every bounded input $u(t) : |u(t)| < M_1 < \infty$, the output of the system stays bounded, that is, $y(t) < M_2 < \infty$.

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Corollary (BIBO Stability and Impulse Response)

a SISO system is BIBO stable if and only if its impulse response $g(t)$ is absolutely integrable in the interval $[0, \infty)$, that is, if

$$\int_0^\infty |g(\tau)|\, d\tau \leq c \tag{2}$$

for some finite constant c.

Corollary (BIBO Stability and Steady State Response)

If a system with transfer (matrix) function $G(s)$ is BIBO stable, then when $t \to \infty$

1. *The output excited by $u(t) = a$, when $t \geq 0$, approaches*

$$G(0)a \tag{3}$$

2. *The output excited by $u(t) = \sin(\omega_0 t)$, when $t \geq 0$, approaches*

$$|G(j\omega_0)| \sin(\omega_0 t + \angle G(j\omega_0)) \tag{4}$$

Sufficient Condition

If every pole of $G(s)$ has negative real part, it is sufficient to conclude that $G(s)$ is BIBO stable.

Definition (Marginal Stability)

For a system, if any finite initial state $x(0)$ excites a bounded response $x(t)$, then it's marginally stable, Lyapunov stable, or just stable.

Corollary (OLHP)

The imaginary axis on the complex plane serves as the stability boundary. A system with poles in the open left-half plane (OLHP) is stable.
If the system transfer function has simple poles that are located on the imaginary axis, it is termed as marginally stable. The impulse response of such systems does not go to zero as $t \to \infty$, but stays bounded in the steady-state.

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Definition (Internal Stability)

The concept of Internal Stability refers to stability of the system response to initial conditions only, assuming zero inputs.

The internal stability is to study the response of the autonomous system (zero input) below

$$\dot{x}(t) = Ax(t), \text{with } x(0) = x_0 \tag{5}$$

Definition (Input-Output Stability)

The Input-Output Stability refers to stability of the response to inputs only, assuming zero initial conditions

It's to study the response of the zero initial state below

$$\dot{x}(t) = Ax(t) + Bu(t), \text{with } x(0) = 0 \tag{6}$$



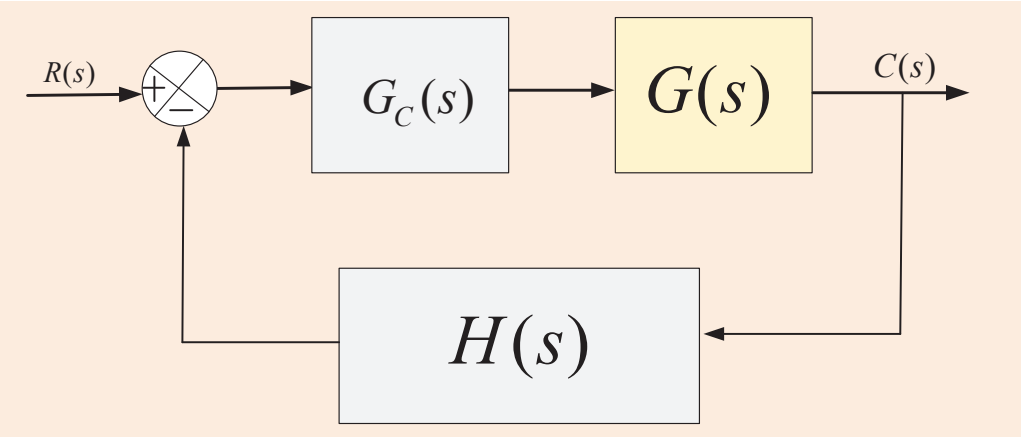
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Suppose the plant transfer function is:

$$G(s) = \frac{1}{s+1} \tag{7}$$

and the controller is given as

$$G_C(s) = K\left(\frac{s+1}{s+10}\right) \tag{8}$$



If $H(s) = 1$, then,

$$G_C(s)G(s) = \frac{K(s+1)}{(s+1)(s+10)} = \frac{K}{s+10} \tag{9}$$

which includes an OLHP pole-zero cancelation. The closed-loop characteristic polynomial:

$$\Delta(s) = s + 10 + K \tag{10}$$

has stable roots for $K > -10$. Hence, the closed-loop system is internally stable for $K > -10$.

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Definition (Asymptotic Stability)

For a general nonlinear system model,

$$\dot{x} = f(x, u) \tag{11}$$

stability of an equilibrium point (x_e, u_e) defined by:

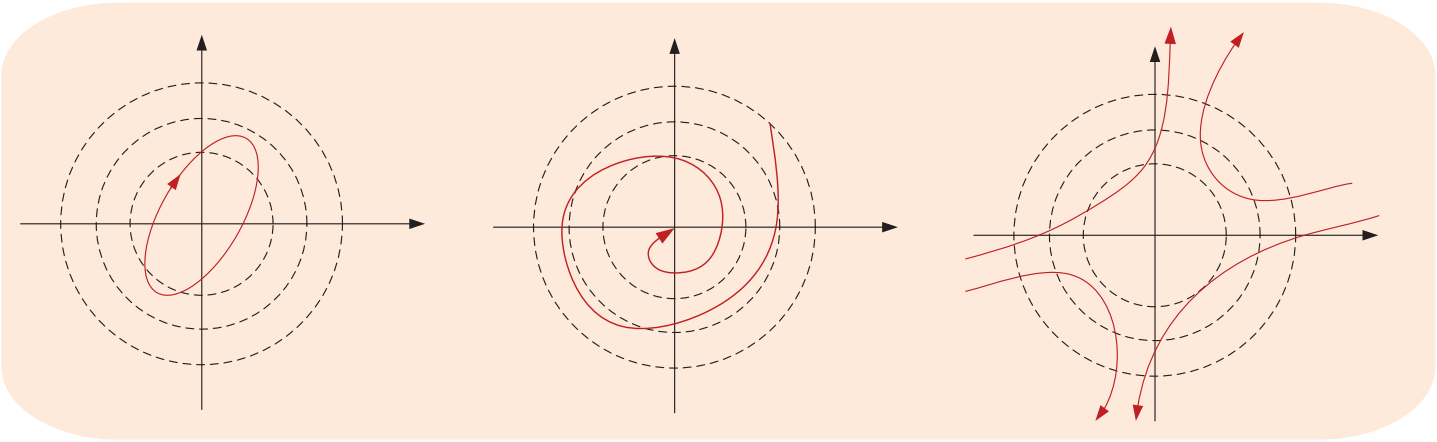
$$f(x_e, u_e) = 0 \tag{12}$$

The equilibrium point is stable (asymptotically stable) if a system trajectory, $x(t)$, that starts in the vicinity of x_e stays close (converges) to x_e. For linear system models, defined by:

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{13}$$

The asymptotic stability requires that the poles of the system transfer function (equivalently, the eigenvalues of the system matrix) lie in the open left-half plane, or $Re(pi) < 0$, and there are no RHP pole-zero cancelations .

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Lyapunov Stable　　Asymptotic Stable　　Unstable

Stability

1. The system will produce a bounded output for every bounded input;
2. If null input, the output should tend to zero, irrespective of any initial conditions.

Stability is the primary requirement for the controller.
The performances might be not so fast, smooth, accurate, but it must be stable.

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Table 1: System Analysis

Method	Model Type	Definition	Math Expression
Routh Criteria	Transfer function, Closed-loop and Open Loop.	The first Column of Routh Array Have the Same Sign.	
Hurwitz Criteria	Transfer function, Closed-loop and Open Loop.	Determination calculation of characteristic function matrix.	
Bode Diagram	Transfer function, Closed-loop and Open Loop.	The logarithm magnitude and the phase angle plots against the frequency.	$20\log\|\|G(j\omega)\|\|, \angle G(j\omega), \omega \nearrow$
Polar Plot	Transfer function, Closed-loop and Open Loop.	The magnitude **v.s.** the phase angle of $G(j\omega)$ on polar coordinates as ω varies from 0 to ∞, also named as Nyquist plot.	$\|G(j\omega)\|\angle G(j\omega)$ $\omega = \begin{cases} 0 \nearrow \infty \\ -\infty \nearrow \infty \end{cases}$
Nyquist Method	Transfer function, Closed-loop	Stability of closed-loop systems from their open-loop frequency response and open-loop poles.	Z= P- 2N or Z= P -N
Root Locus	Transfer function, Closed-loop	The location of the closed-loop poles when the loop gain varies from $-\infty$ to $+\infty$.	$G(s)H(s) = -1$
Lyapunov Criteria	Transfer function, Linear and Nonlinear, Differential and Transfer Equations, Closed-loop and Open-loop Systems	Dissipativity of General Energy	$\dot{V}(x) < 0$ when $V(x) > 0$ or $\dot{V}(x)V(x) < 0$
Robust Stability	Transfer function, Closed-loop and Open-loop Systems, Linear and Nonlinear.	Stability and Ability against Uncertainties.	$\forall\ \Delta \in \Xi, \Re(p_i) \leq 0$

2 Routh and Hurwitz Stability Criteria
- Routh Criteria
- Hurwitz Criteria

If the system transfer function is

$$G(s) = \frac{C(s)}{R(s)} = \frac{M(s)}{D(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \tag{14}$$

whose characteristic function is as follows

$$D(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0 \tag{15}$$

where a_0, a_1, $\cdots$, and a_n are the coefficients of s^n, s^{n-1}, $\cdots$, and s^0, respectively, that $a_i \in \mathcal{R}^1$

Table 2: Routh's Stability Criteria

Order of 's'				
s^n	a_0	a_2	a_4	$\cdots$
s^{n-1}	a_1	a_3	a_5	$\cdots$
s^{n-2}	b_1	b_2	b_3	$\cdots$
s^{n-3}	c_1	c_2	c_3	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
s^0	a_n			



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The parameters in Table.2 are calculated by determinant method below

$$b_1 = \frac{-\begin{vmatrix} a_0 & a_2 \\ a_1 & a_3 \end{vmatrix}}{a_1}, \quad b_2 = \frac{-\begin{vmatrix} a_0 & a_4 \\ a_1 & a_5 \end{vmatrix}}{a_1}, \quad b_3 = \frac{-\begin{vmatrix} a_0 & a_6 \\ a_1 & a_7 \end{vmatrix}}{a_1}, \cdots$$

$$c_1 = \frac{-\begin{vmatrix} a_1 & a_3 \\ b_1 & b_2 \end{vmatrix}}{b_1}, \quad c_2 = \frac{-\begin{vmatrix} a_1 & a_5 \\ b_1 & b_3 \end{vmatrix}}{b_1}, \quad c_3 = \frac{-\begin{vmatrix} a_1 & a_7 \\ b_1 & b_5 \end{vmatrix}}{b_1}, \cdots$$

$$d_1 = \frac{-\begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}}{c_1}, \quad d_2 = \frac{-\begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}}{c_1}, \quad d_3 = \frac{-\begin{vmatrix} b_1 & b_4 \\ c_1 & c_4 \end{vmatrix}}{c_1}, \cdots$$

When the cells of the light gray column in Table.2 have the same sign (positive when a_0 is positive, or negative when a_0 is negative), the system described by the characteristic function in (15) is stable.



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Example

Check the stability of the following systems

$$\frac{s^2+3s-5}{s^3+6s^2+12s+6}=G(s) \tag{16}$$

The characteristic function $D(s)$ above is

$$D(s)=s^3+6s^2+12s+6 \tag{17}$$

whose Routh array is

s^3	1	12
s^2	6	6
s^1	b_1	
s^0	c_1	

$b_1=\dfrac{-\begin{vmatrix}1 & 12\\ 6 & 6\end{vmatrix}}{6}=11, c_1=6$. The values in the first left column are all positive, so we can conclude that the system given in this example is stable .

Example

Check the stability of the following systems

$$\frac{s^3+3s^2+3s-5}{s^4+6s^3+3s^2+s+8}=G(s) \tag{18}$$

The characteristic equation $D(s)$ of above transfer function is

$$D(s)=s^4+6s^3+3s^2+s+8 \tag{19}$$

whose Routh array is

s^4	1	3	8
s^3	6	1	
s^2	b_1	b_2	
s^1	c_1		
s^0	d_1		

$$b_1=\frac{-\begin{vmatrix}1&3\\6&1\end{vmatrix}}{6}=\frac{17}{6},\ b_2=\frac{-\begin{vmatrix}1&8\\6&0\end{vmatrix}}{6}=8,\ c_1=\frac{-\begin{vmatrix}6&1\\\frac{17}{6}&8\end{vmatrix}}{\frac{17}{6}}=-\frac{271}{17},\ d_1=8.\ c_1 \text{ is}$$

Some special cases:

- **One of the first-column term is zero:** If there is zero term in the first-column but others are non-zero, then we can replace the zero-term by a small positive number $\eta \to 0$, $\eta > 0$, substitute the small positive number into the cross-multiplying operation to formulate Routh table.
- **All terms in one derived row are zeros:** In such case, the evaluation of the rest of the array can be continued by forming an auxiliary polynomial with the coefficients of the last row and by using the coefficients of the derivative of the polynomial in the next row.



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Example

Check the stability of the following systems

$$\frac{s^3 + 3s^2 + 3s - 5}{s^4 + 3s^3 + s^2 + 3s + 1} = G(s) \tag{20}$$

The characteristic equation $D(s)$ of above transfer function is

$$D(s) = s^4 + 3s^3 + s^2 + 3s + 1 \tag{21}$$

whose first two rows of the Routh array are

s^4	1	1	1
s^3	3	3	

One can find that the first element of s^2 row would be zero, if the former Routh criteria are utilized. Then, according to the method given for the '**One of the first-column term is zero**' case, one has a positive parameter $\eta \to 0$ that



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Example

s^2	η	1
s^1	c_1	
s^0	1	

where $c_1 = \dfrac{-\begin{vmatrix} 3 & 3 \\ \eta & 1 \end{vmatrix}}{\eta} = -\dfrac{3}{\eta} + 3$

One can find that $\lim_{\eta \to 0} \left(-\dfrac{3}{\eta} + 3 \right) = -\infty$. The system is not stable.



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Example

Check the stability of the following systems

$$\frac{s-5}{s^7+4s^6+5s^5+2s^4+4s^3+16s^2+20s+8}=G(s) \tag{22}$$

The characteristic equation $D(s)$ of above transfer function is

$$D(s)=s^7+4s^6+5s^5+2s^4+4s^3+16s^2+20s+8 \tag{23}$$

whose Routh table is

s^7	1	5	4	20
s^6	4	2	16	8
s^5	$b_1=\frac{9}{2}$	$b_2=0$	$b_3=18$	
s^4	$c_1=2$	$c_2=0$	$c_3=8$	
s^3	0	0		

$$b_1=\frac{-\begin{vmatrix}1&5\\4&2\end{vmatrix}}{4}=\frac{9}{2},\ b_2=\frac{-\begin{vmatrix}1&4\\4&16\end{vmatrix}}{4}=0,\ b_3=\frac{-\begin{vmatrix}1&20\\4&8\end{vmatrix}}{4}=18,$$

Example

$$c_1 = \frac{-\begin{vmatrix} 4 & 2 \\ b_1 & b_2 \end{vmatrix}}{b_1} = 2,\ c_2 = \frac{-\begin{vmatrix} 4 & 16 \\ b_1 & b_3 \end{vmatrix}}{b_1} = 0,\ c_3 = 8.$$

The elements of the s^3 row in the Routh table are all zero, so the auxiliary polynomial is

$$Au(s) = 2s^4 + 8.$$

whose derivative should be

$$\frac{d\left(Au(s)\right)}{ds} = 8s^3 \tag{24}$$

therefore, by defining a positive value $\eta \to 0$, the s^3 row would be continued as

s^4	$c_1 = 2$	$c_2 = 0$	$c_3 = 8$
s^3	8		
s^2	$e_1 = \eta$	$e_2 = 8$	
s^1	$f_1 = -\frac{64}{\eta}$		
s^0	8		

Obviously, $\lim_{\eta \to 0}\left(-\frac{64}{\eta}\right) = -\infty$, system (22) is unstable.

2 Routh and Hurwitz Stability Criteria

- Routh Criteria
- Hurwitz Criteria

Consider the system transfer function

$$G(s) = \frac{C(s)}{R(s)} = \frac{N(s)}{D(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \tag{25}$$

whose characteristic function is as follows

$$D(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0 \tag{26}$$

where a_0, a_1, $\cdots$, and a_n are the coefficients of s^n, s^{n-1}, $\cdots$, and s^0, respectively, that $a_i \in \mathcal{R}^1$.

Theorem

The system (25) is stable, i.e. the roots of the characteristic function (26) have negative real parts, if

$$|\Delta_k| > 0 \tag{27}$$

where $|\Delta_k|$ indicates the determination calculation of matrix $\Delta_k \in \mathcal{R}^{k\times k}$, and

$$\Delta_1 = \Delta_{[1,1]}, \Delta_2 = \Delta_{[1:2,1:2]}, \Delta_3 = \Delta_{[1:3,1:3]}, \ldots, \Delta_k = \Delta_{[1:k,1:k]}, \ldots \Delta_n = \Delta_{[1:n,1:n]} = \Delta$$



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Theorem

$$\Delta_{[1:k,1:k]} = \begin{bmatrix} \delta_{11} & .... & \delta_{1k} \\ \vdots & \vdots & \vdots \\ \delta_{k1} & .... & \delta_{kk} \end{bmatrix} \tag{28}$$

δ_{ij} is the element locating at $i-th$ row and $j-th$ column of the Hurwitz matrix Δ. The Hurwitz matrix function Δ is defined as below

$$\Delta = \begin{bmatrix} a_1 & a_3 & a_5 & \cdots & 0 & 0 \\ a_0 & a_2 & a_4 & \cdots & 0 & 0 \\ 0 & a_1 & a_3 & \cdots & 0 & 0 \\ 0 & a_0 & a_2 & \cdots & 0 & 0 \\ 0 & 0 & a_1 & \cdots & 0 & 0 \\ 0 & 0 & a_0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & 0 & \cdots & a_n & 0 \\ 0 & 0 & 0 & \cdots & a_{n-1} & 0 \\ 0 & 0 & 0 & \cdots & a_{n-2} & a_n \end{bmatrix} \in \mathcal{R}^{n\times n} \tag{29}$$



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3 Bode Diagram
- The Typical Bode Plots
- Phase and Gain Margins
- Minimum Phase System
- Transport Delay



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Definition (Bode Diagram)

For $G(s)$, open or closed-loop, to determine the stability of and performances, the amplitude and phase angle can be approximately calculated with ω varies from 0 to ∞ if s is replaced by $s = j\omega$, as below

$$\begin{aligned} Magnitude: &\quad 20\log|G(j\omega)| \ \ decibels\ dB \\ Phaseangle: &\quad \underline{/G(j\omega)} \end{aligned} \tag{30}$$

Advantage of Bode diagram: the magnitude multiplication is converted into addition .

Proportion Gain

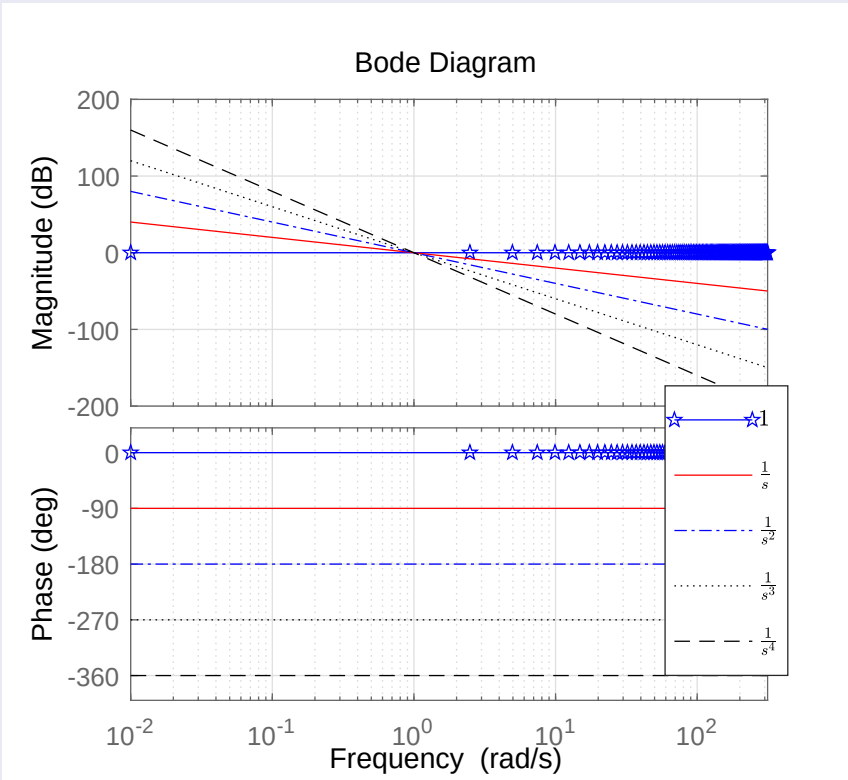
$$20\log|K| = -20\log\frac{1}{|K|} \tag{31}$$



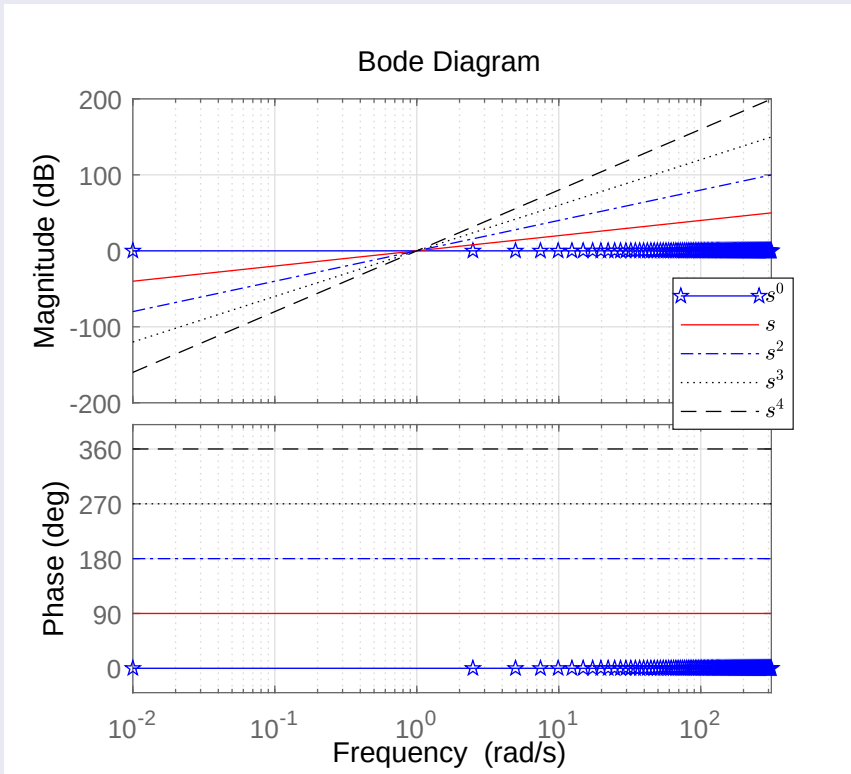
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Integral s^{-n} and Derivative Factor s^n

$$\begin{aligned} 20\log|(j\omega)^n| &= 20n\log|\omega|, \angle(j\omega)^n = \frac{n\cdot\pi}{2} \\ 20\log|(j\omega)^{-n}| &= -20n\log|\omega|, \angle(j\omega)^{-n} = \frac{-n\cdot\pi}{2} \end{aligned} \tag{32}$$



(a) Bode Diagram of $\frac{1}{(j\omega)^n}$



(b) Bode Diagram of $(j\omega)^n$

First-Order Factor $\frac{1}{1+j\omega T}$

The log / decibel magnitude

$$
\begin{aligned}
20\log\left|\frac{1}{1+j\omega T}\right| &= -20\log\sqrt{1+\omega^2T^2} \\
&\approx \begin{cases} 0 & \omega \in (0, \frac{1}{T}] \\ -20\log(\omega T), & \omega \in (\frac{1}{T}, \infty) \end{cases} \quad dB \\
\angle\frac{1}{1+j\omega T} &= -\tan^{-1}\omega T
\end{aligned} \tag{33}
$$

$\omega = \frac{1}{T}$ is the corner frequency.



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The magnitude of

$$G(j\omega) = \frac{1}{1 + 2\eta\left(\frac{j\omega}{\omega_n}\right) + \left(\frac{j\omega}{\omega_n}\right)^2} \tag{34}$$

can be calculated as

$$|G(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\eta^2\frac{\omega^2}{\omega_n^2}}} \tag{35}$$

The frequency ω_r at which $|G(j\omega)|$ reaches its peak is called the resonant frequency , i.e.

$$\omega_r = \arg\max_{\omega} |G(j\omega)| \tag{36}$$

The peak value of $|G(j\omega)|$ occurs when $\omega = \omega_r = \omega_n\sqrt{1 - 2\eta^2}$ for $\eta \in \left[0, \frac{\sqrt{2}}{2}\right]$.



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Damping Ratio Parameter η

- $\eta \to 0$, the resonant frequency ω_r approaches ω_n.
- $0 < \eta \leq \dfrac{\sqrt{2}}{2}$, $\omega_r < \omega_d = \omega_n\sqrt{1-\eta^2}$, ω_d is the damped natural frequency.
- $\dfrac{\sqrt{2}}{2} < \eta$, there is no resonant peak. The magnitude $|G(j\omega)|$ decreases monotonically with increasing frequency ω.

The magnitude of the resonant peak $M_r = |G(j\omega_r)|$ can be calculated as

$$M_r = |G(j\omega)|_{max} = \begin{cases} |G(j\omega_r)| = \dfrac{1}{2\eta\sqrt{1-\eta^2}}, & \text{for} \quad \eta \in [0, \dfrac{\sqrt{2}}{2}] \\ 1, & \text{for} \quad \eta \in (\dfrac{\sqrt{2}}{2}, 1) \end{cases} \tag{37}$$

The phase angle $G(j\omega)$ of the resonant peak occurs at the frequency ω_r can be calculated as

$$\angle G(j\omega_r) = -\tan^{-1}\left(\frac{\sqrt{1-2\eta^2}}{\eta}\right) = -90^\circ + \sin^{-1}\left(\frac{\eta}{\sqrt{1-\eta^2}}\right) \tag{38}$$

The functional relationship, the resonant peak magnitude changes with the damping ratio, is in Fig.1.

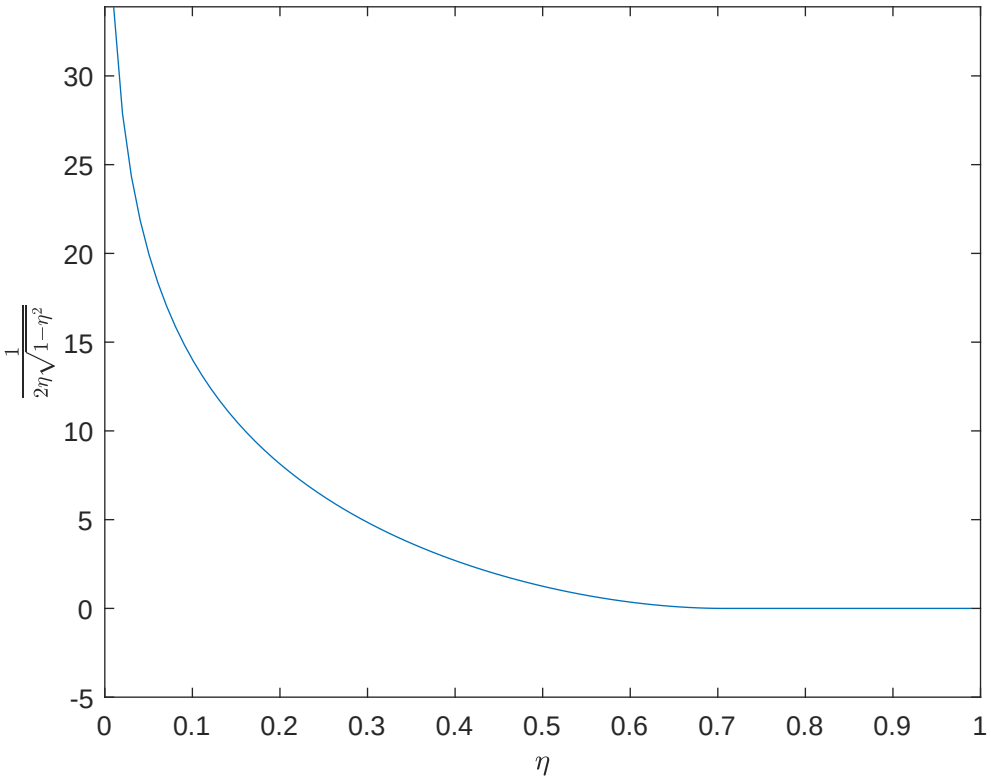


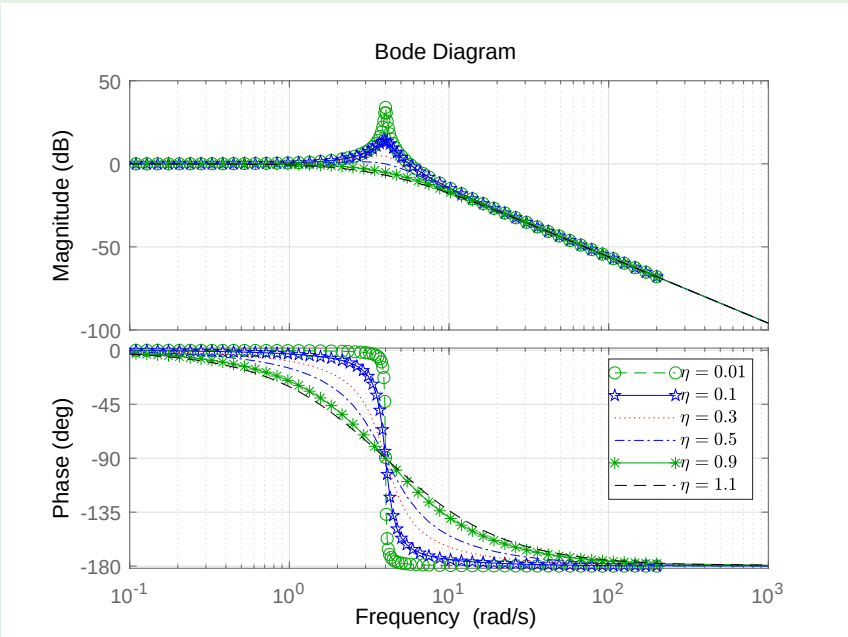
Figure 1: Resonant Peak Value M_r *v.s.* Damping Ratios

Example

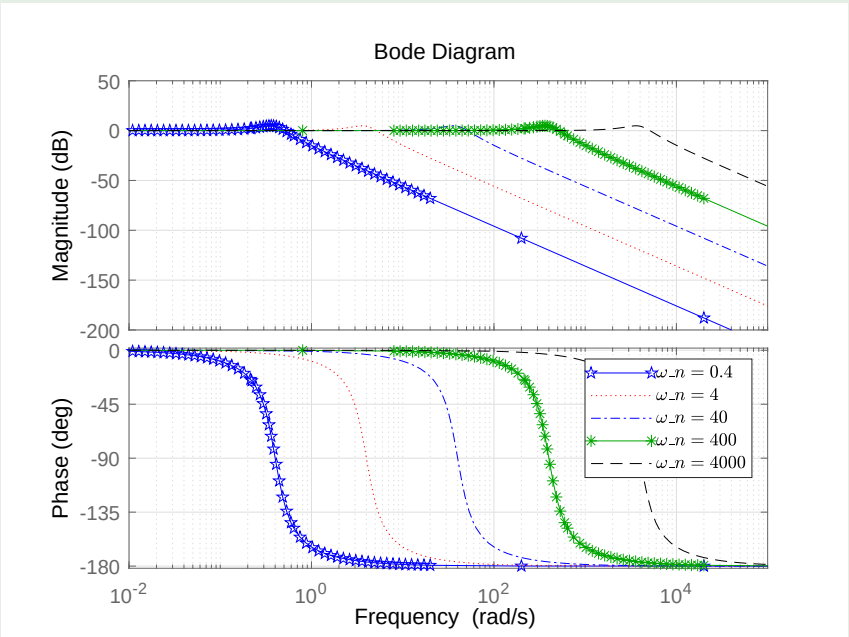
Consider a second order system as follows

$$\frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2} \tag{39}$$

Compare and examine the bode diagrams $\eta = 0.1, 0.3, 0.5, 0.9$ and 1.1 with the same $\omega_n = 4$; the bode plot variations of $\omega_n = 0.4, 4, 40, 400$ and 4000 with the same damping ratio $\eta = 0.3$.



(a) Bode Diagram with Different Damping Ratios



(b) Bode Diagram with Different Corner Frequencies

Comparison

1. different damping ratios η take similar values at the initial and terminal stages;
2. The overshoot magnitudes are higher under the smaller damping ratios η;
3. the log-magnitudes for different corner frequencies ω_n have the same shape, and their phase-frequency curves are also similar;
4. The turning points of the bode diagrams are varied, but they are related with the corresponding corner frequencies;
5. The damping ratio η determines the overshoot magnitude and will lead to the maximal error magnitude with the asymptotic approximations, especially with the low damping ratio values.



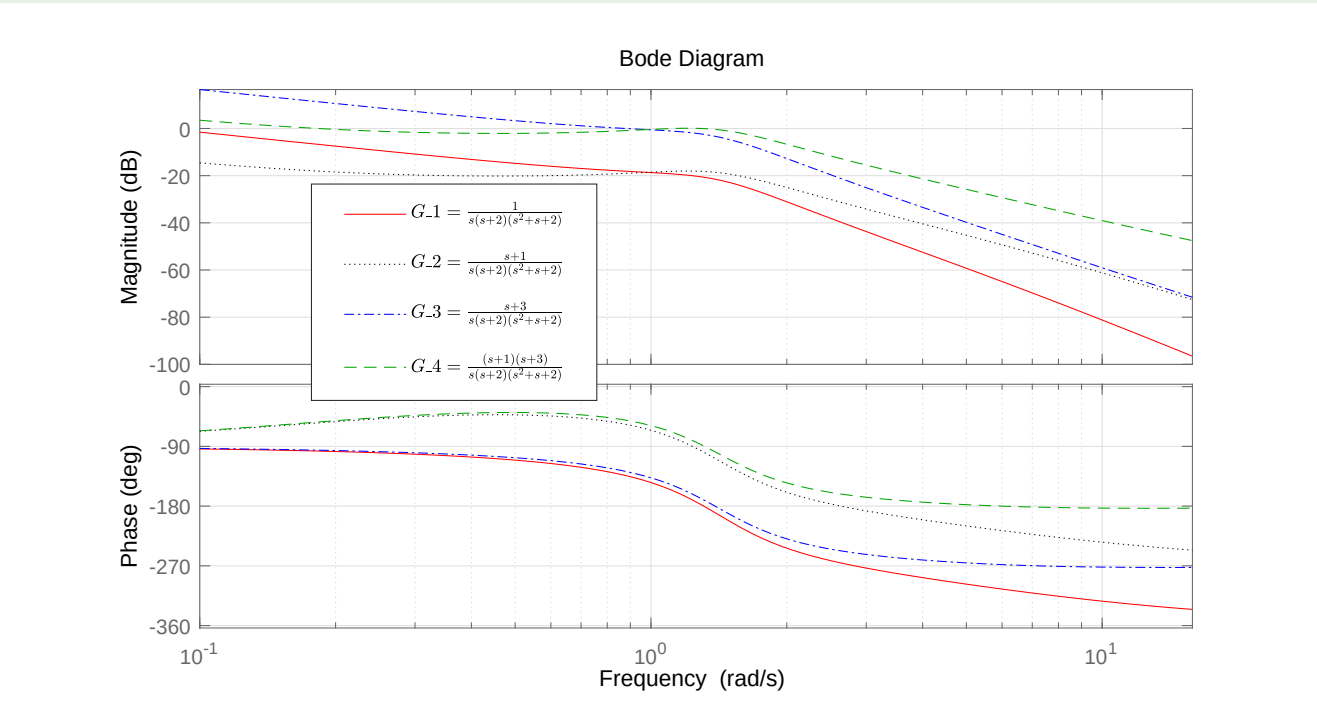
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Example

Draw the Bode diagram of systems $G(s)$ below

$$\begin{aligned} G_1(s) &= \frac{1}{s(s+6)(s^2+s+2)}, G_2(s) = \frac{s+0.2}{s(s+6)(s^2+s+2)} \\ G_3(s) &= \frac{s+8}{s(s+6)(s^2+s+2)}, G_4(s) = \frac{(s+0.2)(s+8)}{s(s+6)(s^2+s+2)} \end{aligned} \tag{40}$$

and compare the effectiveness of poles and zeros.



3 Bode Diagram
- The Typical Bode Plots
- Phase and Gain Margins
- Minimum Phase System
- Transport Delay



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Definition

The gain crossover frequency is the frequency at which $|G(j\omega)|$, the magnitude of the open-loop transfer function, is unity. The phase margin γ is 180° plus the phase angle ϕ of the open-loop transfer function at the gain crossover frequency, or

$$\gamma = 180^\circ + \phi \tag{41}$$

Definition

The gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the frequency at which the phase angle is -180°. Define the phase crossover frequency ω_1 to be the frequency at which the phase angle of the open-loop transfer function equals -180°; then the gain margin K_g is

$$K_g = \frac{1}{|G(j\omega_1)|} \tag{42}$$

In decibels

$$K_g \text{ dB} = 20 \log K_g = -20 \log |G(j\omega_1)| \tag{43}$$



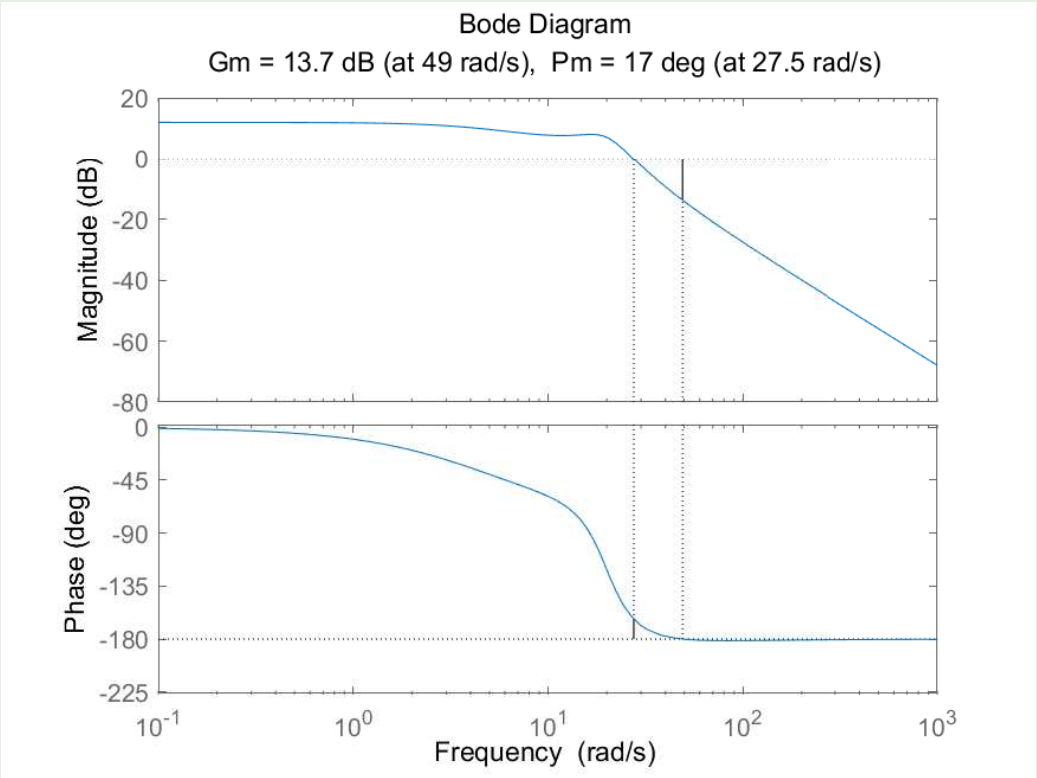
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Example

Consider an open loop system $G(s)$ below

$$G(s) = \frac{100\omega_n^2(s+20)}{(s+5)(s^2+2\eta\omega_n s+\omega_n^2)} \tag{44}$$

where $\omega_n = 20$, $\eta = 0.3$. Calculate the phase and gain margins of system given above.



3 Bode Diagram
- The Typical Bode Plots
- Phase and Gain Margins
- Minimum Phase System
- Transport Delay



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Definition (Minimum Phase System)

A system with all zeros and poles of the transfer function in the left half of the complex plane .

Definition (Non-minimum Phase System)

A system with any zeros and poles of the transfer function in the right half of the complex plane.

A linear, time-invariant system, is said to be minimum-phase if the system and its inverse are causal and stable. When one imposes the constraints of causality and stability, the inverse system is unique.

Example

Determine whether the following open-loop systems are minimum-phase or not

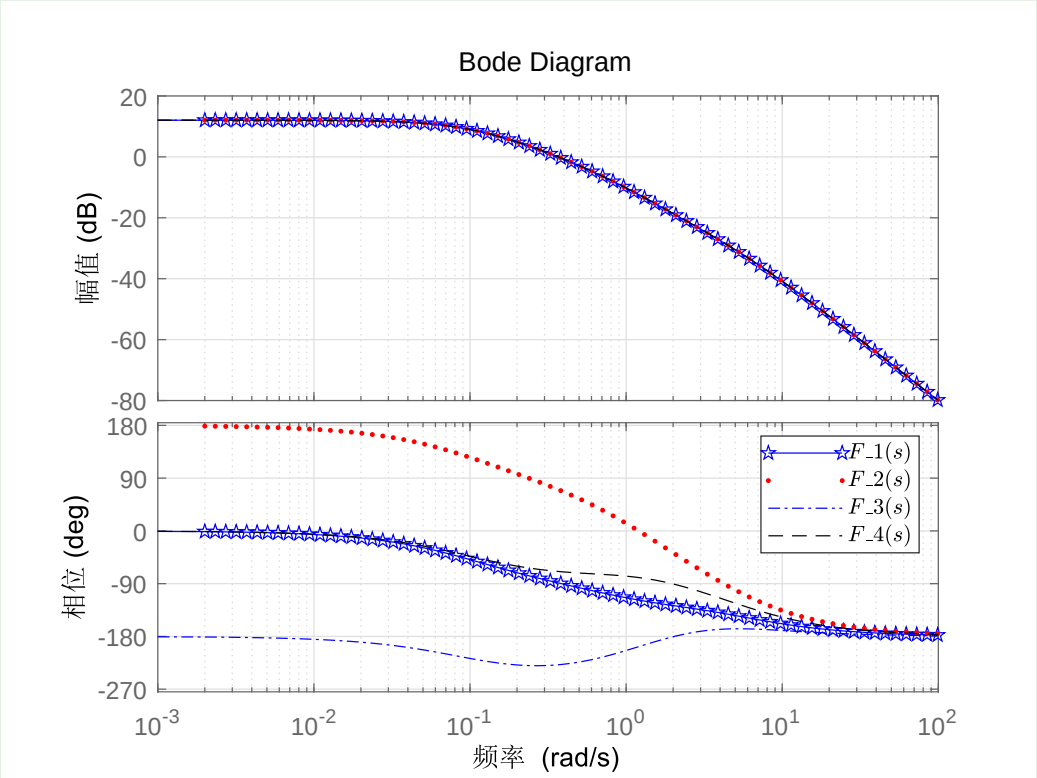
$$\begin{aligned} F_1(s) &= \frac{s+2}{(s+5)(s+1)(s+0.1)}, & F_2(s) &= \frac{s-2}{(s+5)(s+1)(s+0.1)}, \\ F_3(s) &= \frac{s+2}{(s+5)(s-1)(s+0.1)}, & F_4(s) &= \frac{s-2}{(s+5)(s-1)(s+0.1)}. \end{aligned} \tag{45}$$



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Example

$$F_1(s) = \frac{s+2}{(s+5)(s+1)(s+0.1)}, \quad F_2(s) = \frac{s-2}{(s+5)(s+1)(s+0.1)},$$
$$F_3(s) = \frac{s+2}{(s+5)(s-1)(s+0.1)}, \quad F_4(s) = \frac{s-2}{(s+5)(s-1)(s+0.1)}.$$



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3 Bode Diagram
- The Typical Bode Plots
- Phase and Gain Margins
- Minimum Phase System
- Transport Delay



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Transport lags, also called dead times, widely exist in practice

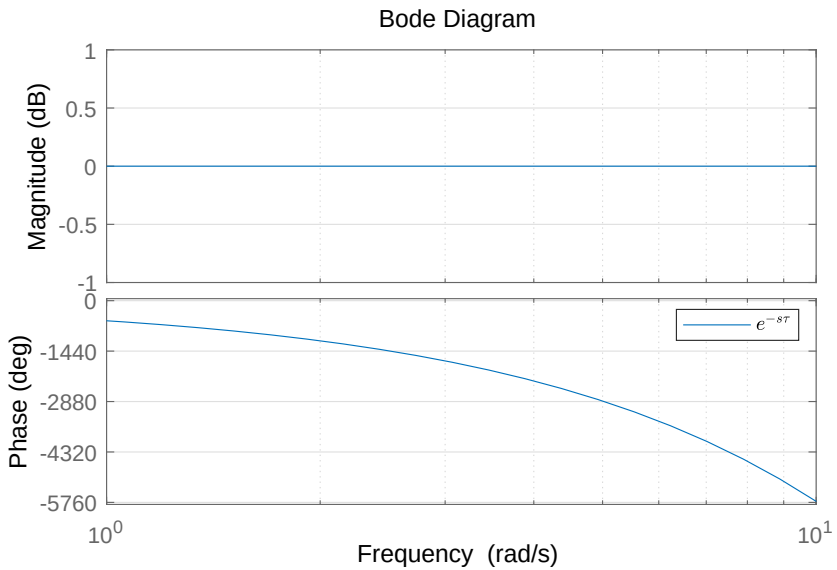
Consider the transport lag given as

$$G(s) = e^{-s\tau} \tag{46}$$

whose magnitude is always unity and 0 dB, and the phase angle of the transport lag is

$$\angle G(j\omega) = -\omega\tau \tag{47}$$

The phase-angle characteristic of transport lag is shown below



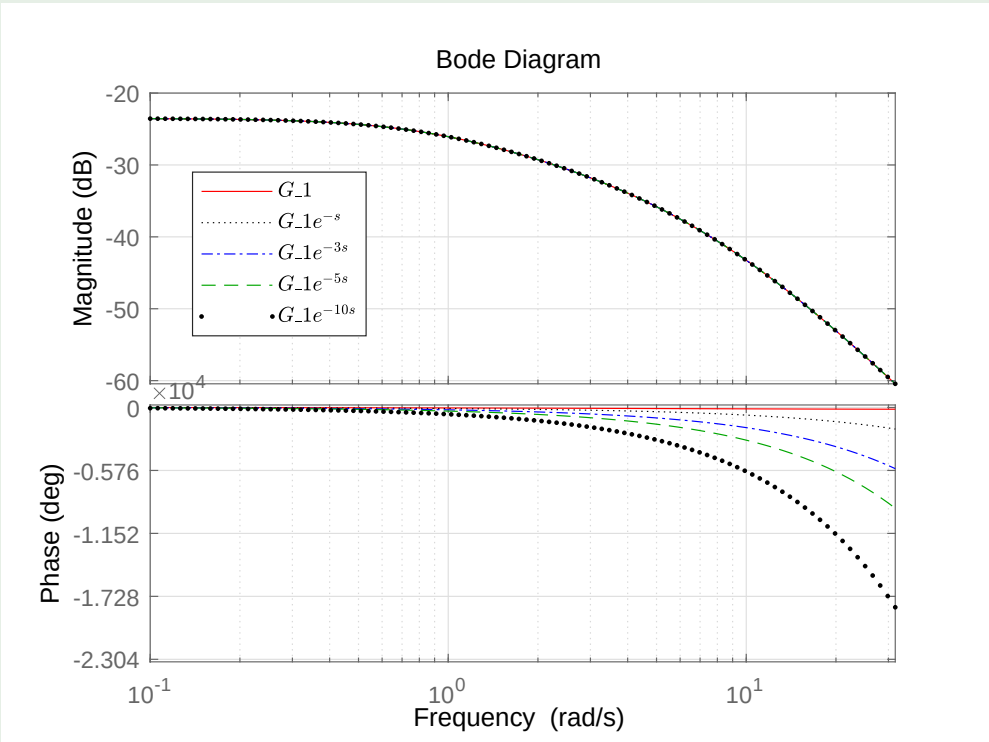
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Example

Compare the bode diagram influence of transport delay of system below

$$G_1(s) = \frac{s+2}{(s+1)(s+3)(s+10)},\ G_2(s) = e^{-s\tau}\frac{s+2}{(s+1)(s+3)(s+10)} \tag{48}$$

where $\tau = 1, 3, 5$ and 10. System comparisons of without transport delay $G_1(s)$ and different delays are



4 Polar Plot
- Polar Plots of First and Second Systems
- Polar Plot of General Systems
- Polar Plots with Time-delay



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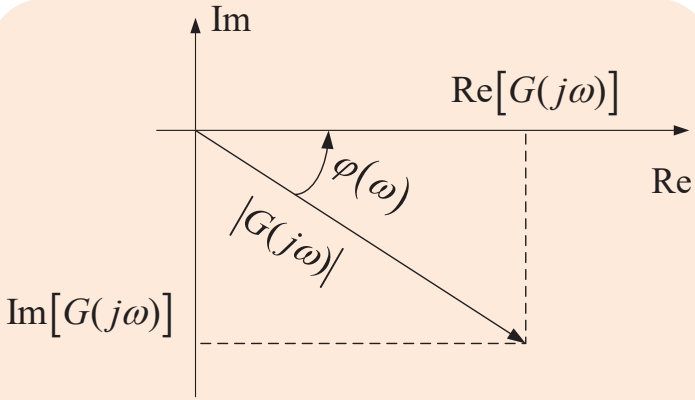
Definition (Polar (Nyquist) plot)

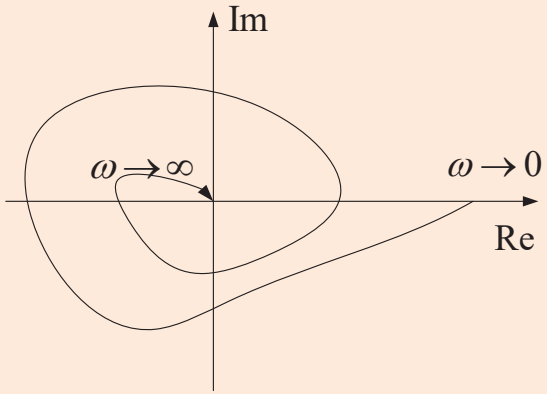
The polar plot of a sinusoidal transfer function $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as ω varies from 0 to ∞.

How to plot

The polar plot is the locus of vectors $|G(j\omega)|\angle G(j\omega)$ with ω changes $0 \nearrow \infty$. The positive (negative) phase angle is measured counterclockwise (clockwise) from the positive real axis.

$$G(j\omega) = |G(j\omega)|\, e^{j\varphi(\omega)} = \mathbf{Re}[G(j\omega)] + j\mathbf{Im}[G(j\omega)] = \|G(j\omega)\|\cos(\varphi(\omega)) + j\|G(j\omega)\|\sin(\varphi(\omega)) \quad (49)$$





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Plotting Steps

- Step 1: Determine the starting $\omega = 0_+$ and end point $\omega = \infty$ of the open-loop amplitude-phase curve

$$\begin{aligned} G(j\omega) &= K\frac{\prod_{i=1}^{m}(jT_i\omega+1)}{(j\omega)^{\nu}\prod_{j=1}^{n-\nu}(jT_j\omega+1)} \\ &= \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_1 j\omega + b_0}{a_n(j\omega)^n + a_{n-1}(j\omega)^{n-1} + \cdots + a_1 j\omega + a_0} \end{aligned}$$

One has the magnitude and phase that

$$\begin{aligned} |G(j\omega)| &= |K|\frac{\prod_{i=1}^{m}\sqrt{1+(T_i\omega)^2}}{(\omega)^{\nu}\prod_{j=1}^{n-\nu}\sqrt{1+(T_j\omega)^2}} \\ \varphi(\omega) &= \sum_{i=1}^{m}\arctan T_i\omega - \nu\frac{\pi}{2} - \sum_{j=1}^{n-\nu}\arctan T_j\omega \end{aligned} \tag{50}$$

Therefore, when $\omega = 0_+$ and $\omega = \infty$ the start phase and the end phase are

$$\begin{aligned} \lim_{\omega\to 0_+}\varphi(\omega) &= -\nu\frac{\pi}{2} \\ \lim_{\omega\to\infty}\varphi(\omega) &= -(n-m)\frac{\pi}{2} \end{aligned} \tag{51}$$

Plotting Steps

- Step 2: Determine the starting and ending magnitudes. The starting magnitude is

$$\lim_{\omega \to 0} |G(j\omega)| = \begin{cases} |K| & when \quad \nu = 0 \\ \infty & when \quad \nu \neq 0 \end{cases} \tag{52}$$

The ending magnitude is

$$\lim_{\omega \to \infty} |G(j\omega)| = 0 \tag{53}$$

- Step 3: Determine intersection point on the real axis. Set

$$\mathbf{Im}[G(j\omega)] = 0 \tag{54}$$

one can obtain the intersection frequencies, ω_1,...,ω_s, and get the intersection points on the real axis that

$$\mathbf{Re}[G(j\omega)]|_{\omega=\omega_i} \tag{55}$$



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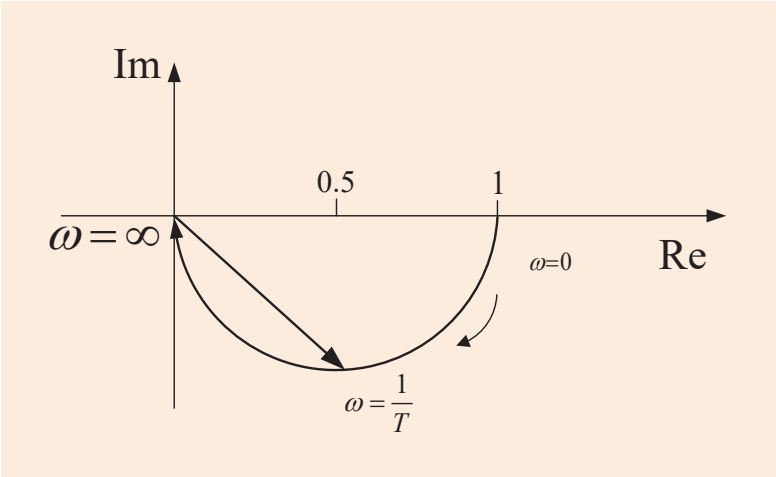
For the first-order system $\frac{1}{1+Ts}$, whose sinusoidal transfer function is

$$G(j\omega) = \frac{1}{1+j\omega T} = \frac{1}{\sqrt{1+\omega^2 T^2}} \angle -\tan^{-1}(\omega T) \tag{56}$$

the values of $G(j\omega)$ at $\omega = 0$ and $\omega = \frac{1}{T}$ are respectively,

$$G(j0) = 1\angle 0^\circ \quad \text{and} \quad G\left(\frac{1}{T}\right) = \frac{1}{\sqrt{2}} \angle -45^\circ \tag{57}$$

When ω goes closely to ∞, the magnitude and phase of $G(j\omega)$ go towards 0 and -90°.



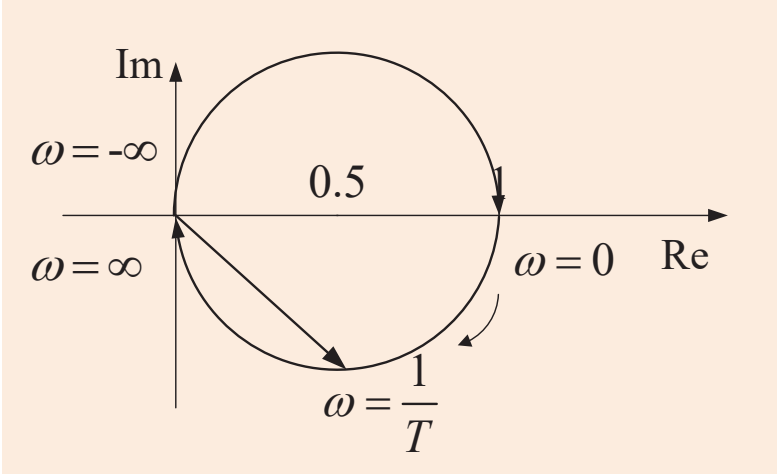


Figure 2: Polar Plot of First-order System

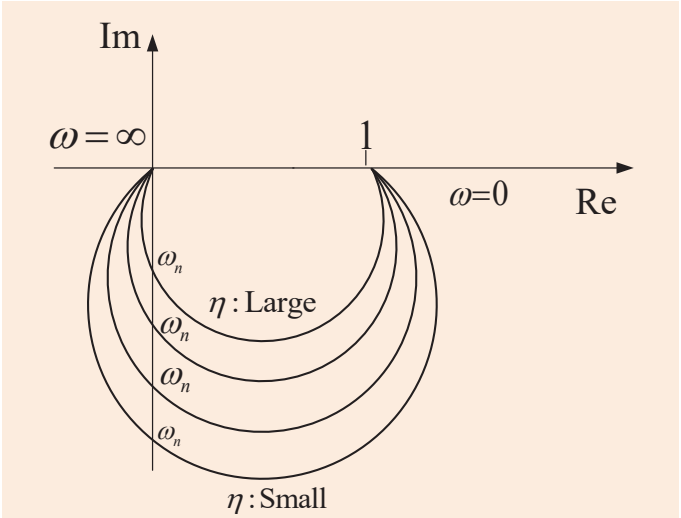
The transfer function of the typical second-order system

$$G(s) = \frac{1}{\frac{s^2}{\omega_n^2} + 2\eta\frac{s}{\omega_n} + 1} = \frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2}, \tag{58}$$

where the parameter $\eta > 0$, whose polar plot can be verified via setting $s = j\omega$ and increasing ω from 0 to ∞, such that

$$\lim_{\omega\to 0} G(j\omega) = 1\angle 0^\circ \text{ and } \lim_{\omega\to\infty} G(j\omega) = 0\angle -180^\circ \tag{59}$$

and the cross point to negative Im axis is $\omega = \omega_n$. The polar plot of this quadratic system depends on the value of the damping ratio η, but the general shape of the plot is the same for both the under-damped case $(1 > \eta > 0)$ and over-damped case $(\eta > 1)$.

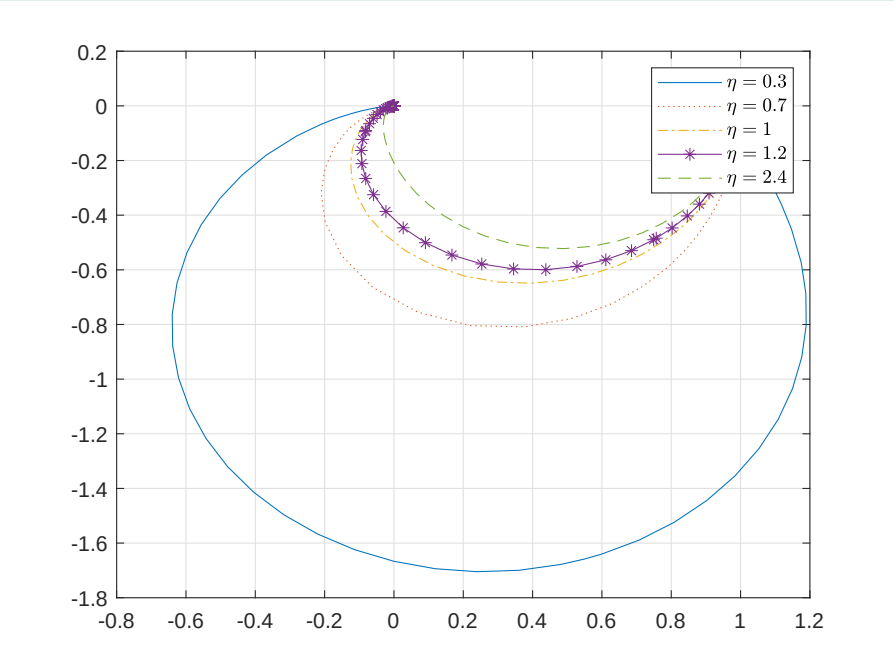


Example

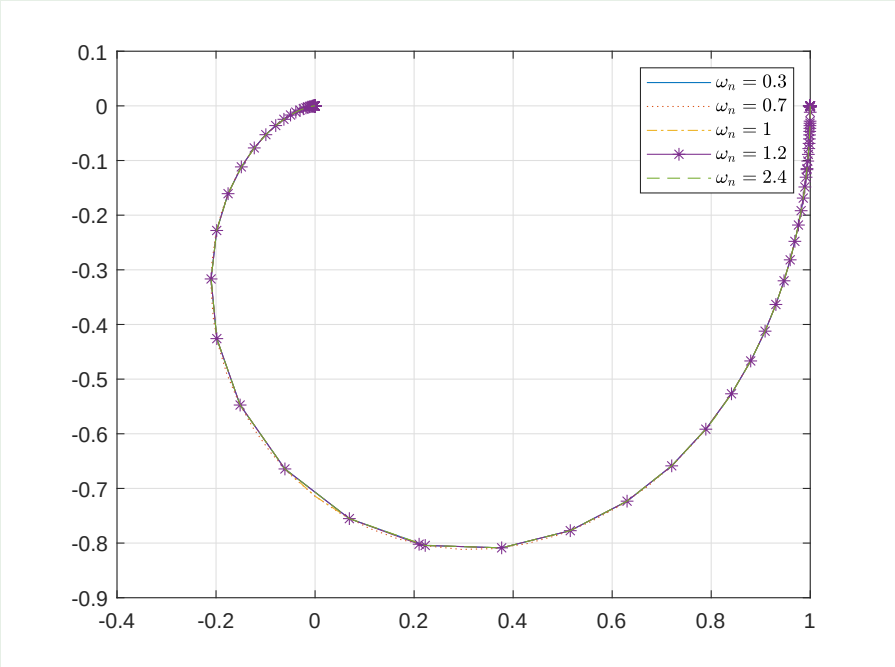
For the second order system,

$$\frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2} \tag{60}$$

Compare and examine the polar plot variations of $\eta = 0.3, 0.7, 1, 1.2$ and 2.4 with the same $\omega_n = 2$; And the polar plot variations of $\omega_n = 0.3, 0.7, 1, 1.2$ and $\eta = 0.7$.



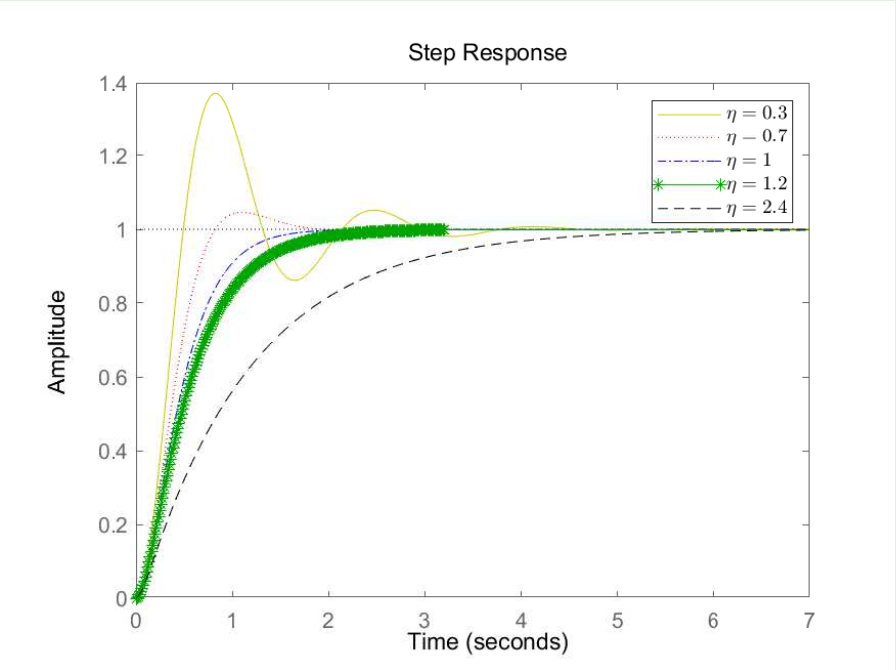
(a) Plots with Different Damping Ratios η



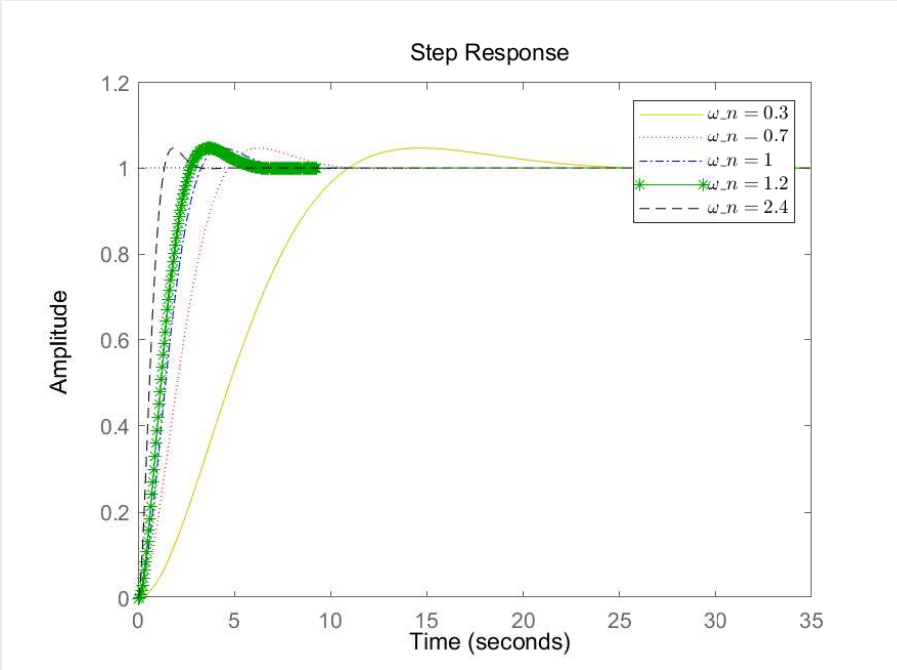
(b) Plots with Different Frequency ω_n

Example

Left Figure: the same natural frequency ω_n but different damping ratios η;
Right Figure: different natural frequencies ω_n but the same damping ratio η;
Though the step responses have the same overshoot amplitude but different overshoot times, their polar plots are the same.



(c) Step Responses with Different Damping Ratios η



(d) Step Responses with Different Frequency ω_n

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4 Polar Plot

- Polar Plots of First and Second Systems
- Polar Plot of General Systems
- Polar Plots with Time-delay

Not considering the time-delay, the general rational system model is

$$\begin{aligned} G(s) &= \frac{K(1+sT_a)(1+sT_b)\cdots}{s^{\lambda}(1+sT_1)(1+sT_2)\cdots} \\ &= \frac{b_0 s^m + b_1 s^{m-1} + \cdots}{a_0 s^n + a_1 s^{n-1} + \cdots} \end{aligned} \tag{61}$$

where $n > m$, i.e. the degree of the denominator polynomial is larger than that of the numerator. Then, the shapes of the polar plots are classified as follows

Type 0 systems, $\lambda = 0$

- The starting point, corresponding to $\omega = 0$, is finite and on the positive real axis.
- The tangent to the polar plot at $\omega = 0$ is perpendicular to the real axis.
- The terminal point, which corresponds to $\omega = \infty$, is at the origin, and the curve is tangent to one of the axes.



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✠ Type 1 systems, $\lambda = 1$

- The $j\omega$ term in the denominator contributes -90° to the total phase angle of $G(j\omega)$ for $0 \leq \omega \leq \infty$.
- At $\omega = 0$, the magnitude of $G(j\omega)$ is ∞, and the phase angle becomes -90°.
- At low frequencies, the polar plot is asymptotic to a line parallel to the negative imaginary axis.
- At $\omega = \infty$, the magnitude becomes zero, and the curve converges to the origin and its tangent to one of the axes.

✠ Type 2 systems, $\lambda = 2$

- The $(j\omega)^2$ term in the denominator contributes -180° to the total phase angle of $G(j\omega)$ for $0 \leq \omega \leq \infty$.
- At $\omega = 0$, the magnitude of $G(j\omega)$ is infinity, and the phase angle is equal to -180°.
- At low frequencies, the polar plot may be asymptotic to the negative real axis.
- At $\omega = \infty$, the magnitude becomes zero, and the curve is tangent to one of the axes.



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Example

Consider a nonminimum phase system model $G(s)$ below

$$G(s) = \frac{s+1}{s(s-5)(s-3)} \tag{62}$$

Replace s with $j\omega$, one has

$$\begin{aligned} G(j\omega) &= \frac{j\omega+1}{j\omega(-\omega^2-8j\omega+15)} \\ &= \frac{8\omega^2+\omega^2(15-\omega^2)-j\omega(15-9\omega^2)}{(8\omega^2+j\omega(15-\omega^2))(8\omega^2-j\omega(15-\omega^2))} \end{aligned} \tag{63}$$

When $\omega^2 = \frac{15}{9}$, the imaginary part of the above equation is zero. Then, the frequency across the real axis is calculated as $\omega^2 = \frac{5}{3}$ and the cross point is $\left(\frac{3}{40}, 0\right)$. And, when $\omega^2 = 23$, the real part is zero, the cross point at the imaginary axis is $\left(0, \frac{\sqrt{23}}{184}j\right)$.



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Example

The system magnitude is

$$|G(j\omega)| = \begin{cases} \dfrac{\sqrt{\omega^2+1}}{|\omega|\sqrt{\omega^2+25}\sqrt{\omega^2+9}}, & \omega \neq 0 \\ \infty, & \omega = 0 \end{cases}$$

and

$$\begin{aligned} \underline{/G(j\omega)} &= -\frac{\pi}{2} + \arctan\omega - \left(\pi - \arctan\frac{\omega}{5}\right) - \left(\pi - \arctan\frac{\omega}{3}\right) \\ &= -\frac{5\pi}{2} + \arctan\omega + \arctan\frac{\omega}{5} + \arctan\frac{\omega}{3} \end{aligned}$$

When $\omega = 0$, remove all of the $'s'$ in system model, one has that

$$|G(j\omega)|_{\omega=0} = \infty, \underline{/G(j\omega)}_{\omega=0} = -\frac{5\pi}{2}$$

When $\omega = 0^+$ and $\omega = \infty$, one has that

$$\begin{aligned} |G(j\omega)|_{\omega=0^+} &= \infty, \underline{/G(j\omega)}_{\omega=0^+} = -\frac{5\pi}{2}, \\ |G(j\omega)|_{\omega=\infty} &= 0, \underline{/G(j\omega)}_{\omega=\infty} = -\pi \end{aligned}$$

4 Polar Plot

- Polar Plots of First and Second Systems
- Polar Plot of General Systems
- Polar Plots with Time-delay

Consider a general shape system with time delay as below

$$G(s) = G_o(s)e^{-s\tau} \tag{64}$$

where $G_o(s)$ is the known original system. Therefore, the system magnitude is

$$|G(j\omega)| = \left|e^{-j\omega\tau}\right| |G_o(j\omega)| \tag{65}$$

and the phase is

$$\angle G(j\omega) = \angle e^{-j\omega\tau} + \angle G_o(j\omega) = -\omega\tau + \angle G_o(j\omega) \tag{66}$$



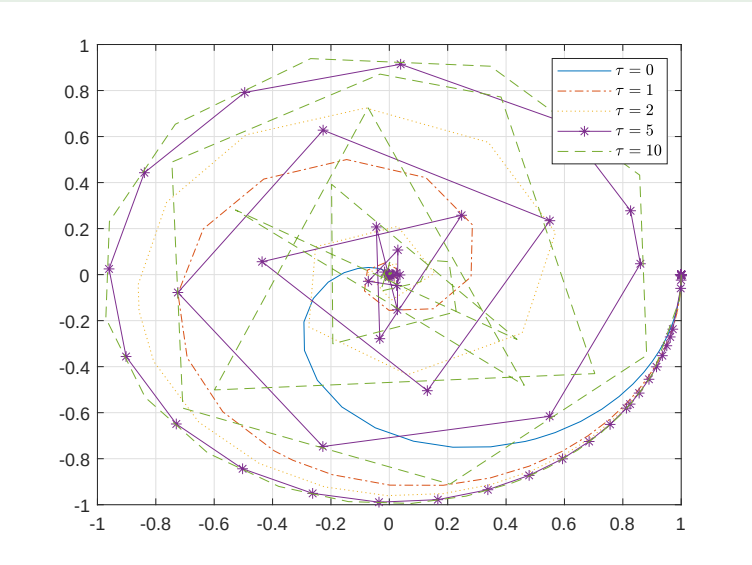
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Example

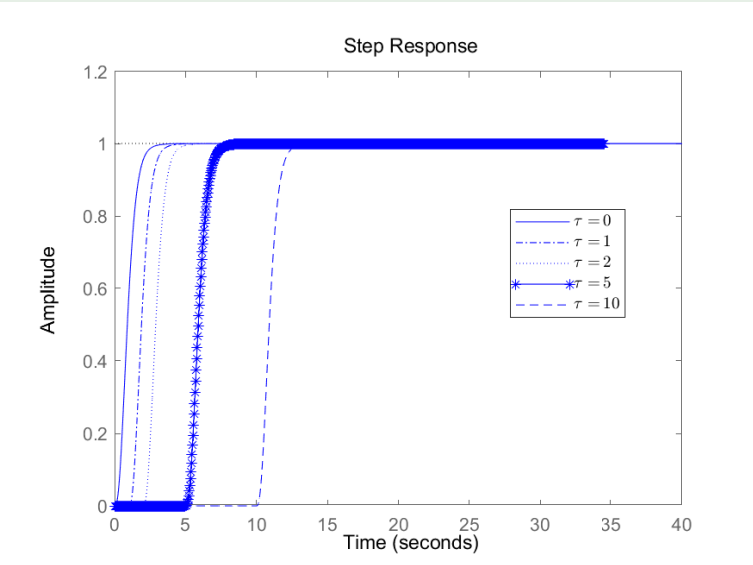
Consider the original system $G_o(s)$ below

$$G_o(s) = \frac{K}{(s+z_1)(s+z_2)(s+z_3)} \tag{67}$$

where $K = 20.5$, $z_1 = 2$, $z_2 = 2.5 + 2j$, and $z_3 = 2.5 - 2j$. Compare the polar plots when term $\tau = 1, 2, 5, 10$ in the delay component $e^{-s\tau}$.



(e) Polar Plots with Different Delays

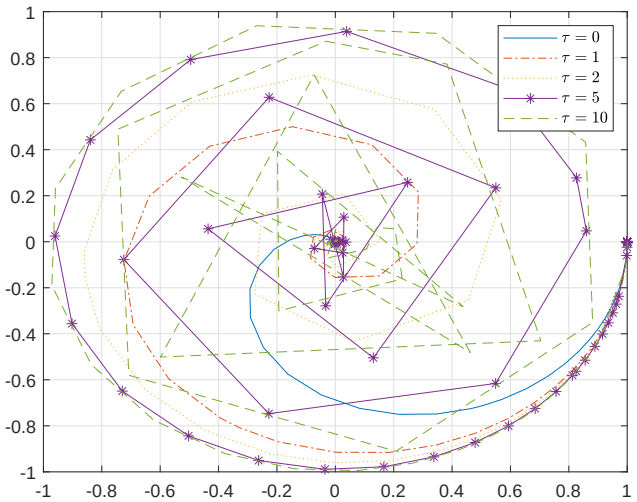


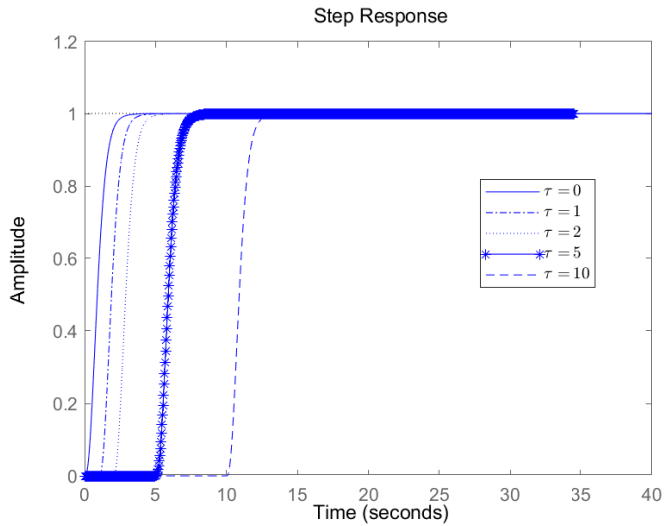
(f) Step Responses with Different Delays

Example

The effects of $G_o(s)e^{-\tau s}$

The time delay component doesn't change the curve shape but postpones the responses by τ. The shapes of the polar plots are more coiled with the larger time delay





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1 Introduction

2 Routh and Hurwitz Stability Criteria

3 Bode Diagram

4 Polar Plot

5 Nyquist Stability Criteria

6 Root Locus Method

7 Lyapunov Stability Criteria

8 Robust Stability

9 Chapter Conclusion

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Why Nyquist Criteria

The Nyquist stability criterion is used to examine the stability of a closed-loop system from its open-loop frequency response and open-loop poles.

For a plant $G(s)$ and its negative feedback block $H(s)$ in Fig.3, the closed-loop system can be

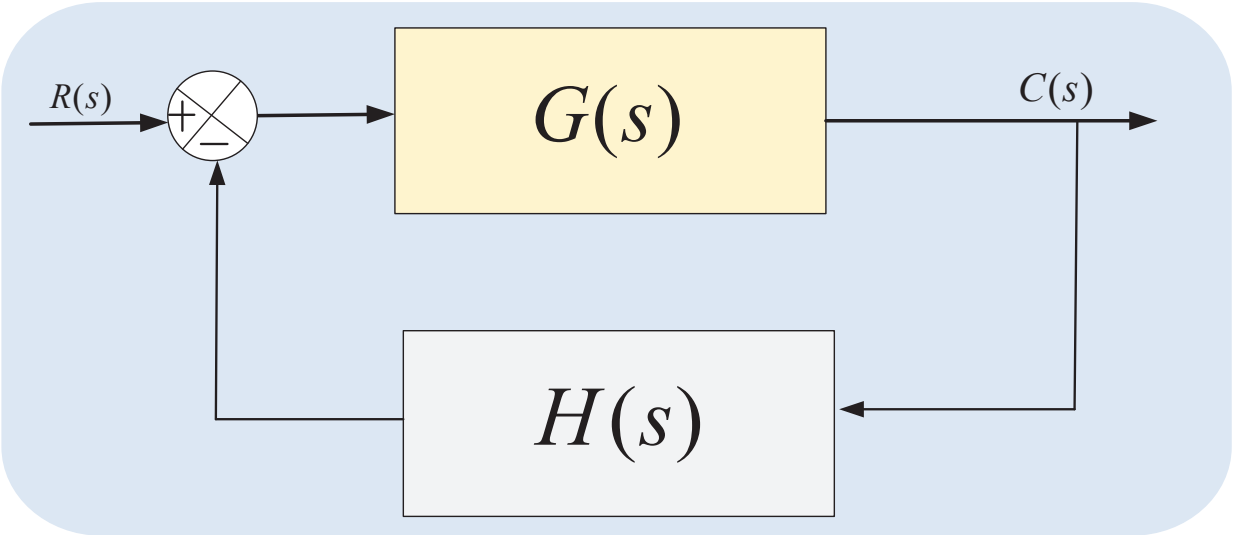


Figure 3: Typical Closed-loop System

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Then the closed-loop system is

$$\Phi(s) = \frac{G(s)}{1 + G(s)H(s)} \tag{68}$$

the open-loop system is

$$F(s) = G(s)H(s) = \frac{K^* N(s)}{D(s)} = \frac{K^*(n_1 s + 1)(n_2 s + 1)\ldots(n_N s + 1)}{(d_1 s + 1)(d_2 s + 1)\ldots(d_D s + 1)} \tag{69}$$

Then one can check whether the equation

$$F(j\omega) = G(j\omega)\, H(j\omega) \tag{70}$$

has a solution on the imaginary axis.

If ω_0 satisfies $1 + F(j\omega_0) = 0$, namely $F(j\omega_0) = -1$, in the closed-loop system, an un-damped oscillation arises, the system gets to the borderline of stability. The Nyquist diagram of the open-loop goes through the point $-1 + j0$ of the complex plane.



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Theorem (Nyquist Criterion)

If Z, the number of closed-loop poles in the right-half s plane, which can be calculated as

$$Z = P - 2N = 0 \tag{71}$$

one can determine that system is stable.

N: the number of clockwise encirclements around the point $-1 + 0j$
P: the number of poles of $G(s)H(s)$ in the right-half s plane.

Example

An open loop transfer function of a state feedback system is shown as follows,

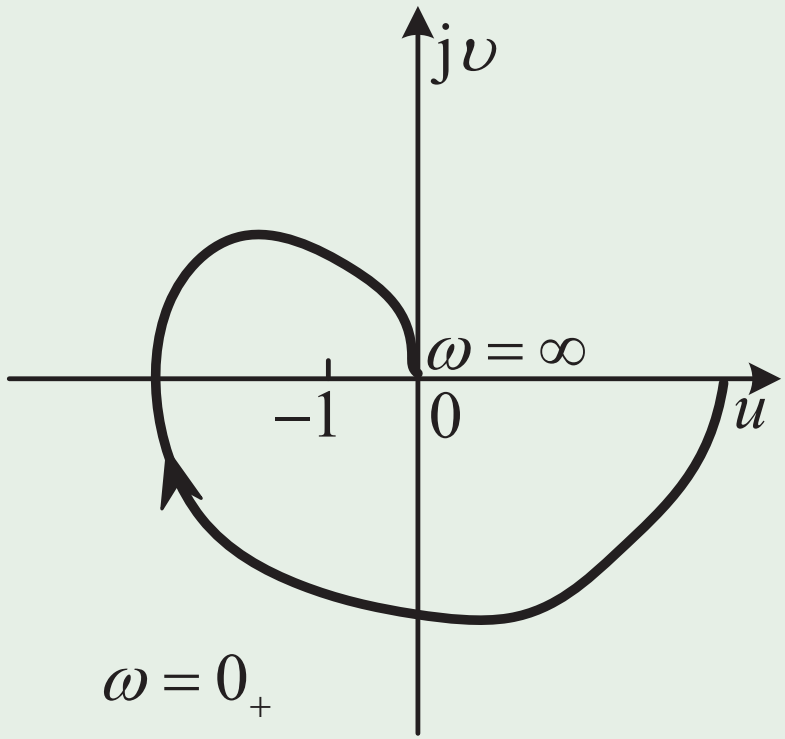
$$G(s)H(s) = \frac{20}{(3s+1)(2s+1)(s+1)} \tag{72}$$

use Nyquist stability criterion to determine the stability of this system.



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Example



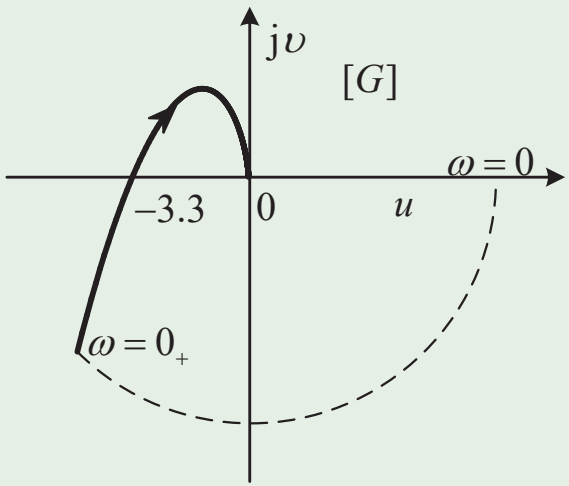
$$Z = P - 2N = 0 - (-2) = 2 \tag{73}$$

Unstable?

Example

An open loop transfer function of a state feedback system is shown as follows, use Nyquist stability method to determine the stability of this closed-loop system

$$G(s)H(s) = \frac{5}{s(2s+1)(s+1)} \tag{74}$$



Then, one has that

$$Z = P - 2N = 0 - (-2) = 2 \tag{75}$$

6 Root Locus Method
- Root Locus Plot
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- Root Locus Plot based Controller Design

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What the Root Locus Method can Do?

Predict and determine the effects on the location of the closed-loop poles of varying the gain value K_p or adding open-loop poles and/or open-loop zeros.

Consider the negative feedback system

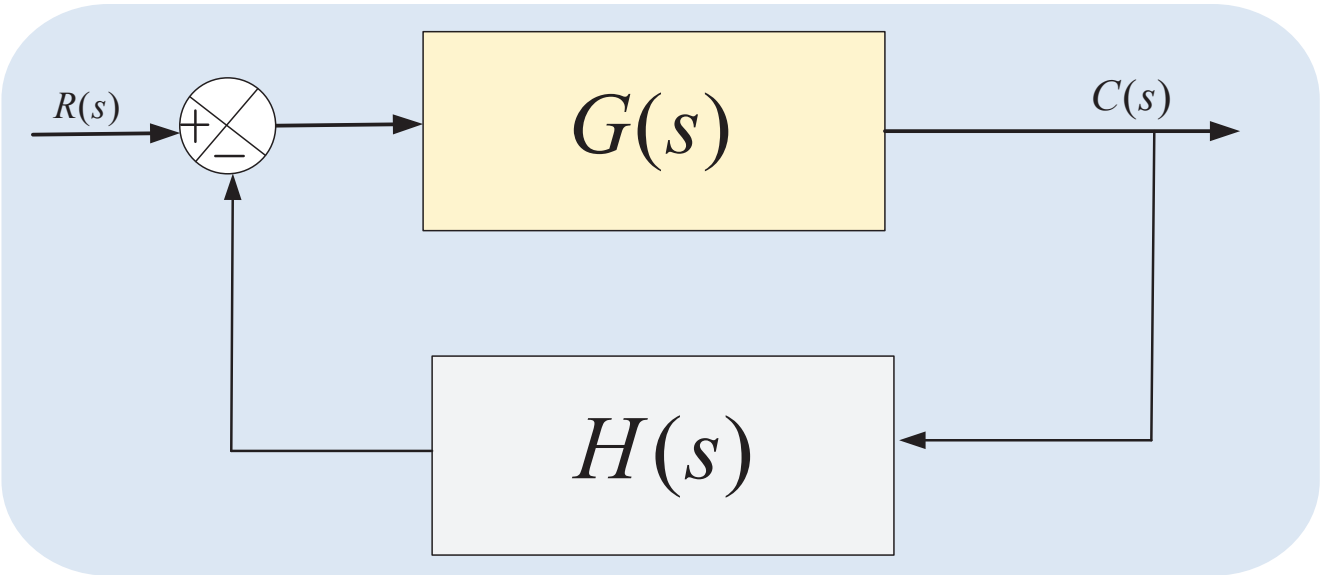


Figure 4: Plant with Negative Feedback

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The closed-loop model is

$$\Phi(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \tag{76}$$

whose characteristic equation is

$$1 + G(s)H(s) = 0 \tag{77}$$

or equally

$$G(s)H(s) = -1 \;\Leftrightarrow\; |G(s)H(s)| = 1,\; \underline{\angle G(s)H(s)} = \pm\pi(2k+1) \tag{78}$$

where

$$\begin{cases} \underline{\angle G(s)H(s)} &= \pm\pi(2k+1), \quad (k = 0, 1, 2, \cdots), \\ |G(s)H(s)| &= 1 \end{cases}$$



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The closed-loop system's characteristic function can also be expressed as

$$F(s) = 1 + G(s)H(s) = 1 + K\frac{(s+z_1)(s+z_2)\cdot(s+z_m)}{(s+p_1)(s+p_2)\cdot(s+p_n)} = 0 \tag{79}$$

The root locus method is to see how $F(s)$ varies with the change of K from $-\infty$ to $+\infty$. Normally K is chosen

$$K \in (-\infty, +\infty) \tag{80}$$

For negative feedback case, one considers that

$$K \in (0, +\infty) \tag{81}$$



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Root-locus Plot Steps for Negative Feedback Systems

- Step 1: Locate the poles and zeros of $G(s)H(s)$ on the s plane.
  - The root-locus branches *start* from the poles and *terminate* at the zeros (finite zeros or zeros at infinity) of the open loop system $G(s)H(s)$.
  - The poles and zeros locate and are determined from the factored form of the open loop transfer function.
  - The number of branches equals that of poles.
- Step 2: Determine the root loci on the real axis by open-loop poles and zeros.
  Root loci on the real axis are determined by open-loop poles and zeros lying on it.
- Step 3: Determine the asymptotes of root loci.

$$\text{Angles of asymptotes} = \frac{\pm\,\pi(2k+1)}{n-m}, \quad (k = 0, 1, 2, \cdots) \tag{82}$$

- Step 4: Find the breakaway and break-in points. Let $G(s)H(s) = \dfrac{KA(s)}{B(s)}$, the characteristic equation can be given

$$B(s) + KA(s) = 0 \tag{83}$$

The breakaway and break-in points can be determined from the roots of

$$\frac{dK}{ds} = -\frac{B'(s)A(s) - B(s)A'(s)}{A^2(s)} = 0 \tag{84}$$

Root-locus Plot Steps for Negative Feedback Systems

- Step 5: Determine the angle of departure.
- Step 6: Find the points where the root loci may cross the imaginary axis.
- Step 7: Taking a series of test points in the broad neighborhood of the origin of the s plane, sketch the root loci.
- Step 8: Determine closed-loop poles. The value of K corresponding to any point s on a root locus can be obtained using the magnitude condition, or

$$K = \frac{\text{product of lengths between point } s \text{ to poles}}{\text{product of lengths between point } s \text{ to zeros}} \tag{85}$$



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Example

Consider the negative feedback system as in Fig.4, i.e. $K \geq 0$, which is

$$G(s) = \frac{K}{s(s+1)(s+2)}, \quad H(s) = 1 \tag{86}$$

sketch the root-locus plot and determine the value K that the damping ratio of closed-loop system $\eta = 0.5$ of a pair of dominant complex-conjugate poles. For the given system, the angle condition is

$$\begin{aligned} \angle G(s) &= \angle \frac{K}{s(s+1)(s+2)} \\ &= \pm\pi(2k+1), \text{for } (k = 0, 1, 2, \cdots) \end{aligned} \tag{87}$$

and the magnitude condition is

$$|G(s)| = \left| \frac{K}{s(s+1)(s+2)} \right| = 1 \tag{88}$$

Example

1. Determine the root loci on the real axis.

- Locate the open-loop poles when $s = 0$, $s = -1$ and $s = -2$.
- The start points of the root loci, i.e. when $K = 0$, are the open-loop poles.
- There are three open-loop poles, therefore, three individual root loci.

When $s \in (-1, 0)$ and $s \in (-\infty, -2)$ the phase angle sum from the three poles to the test point

$$\sum_{i=1}^{3} \angle(s - p_i) = \angle s + \angle(s + 1) + \angle(s + 2) = \pi \tag{89}$$

$$\sum_{i=1}^{3} \angle(s - p_i) = \angle s + \angle(s + 1) + \angle(s + 2) = 3\pi \tag{90}$$

Phase condition for the test point in the ranges $(-1, 0)$ and $(-\infty, -2)$ can be satisfied.

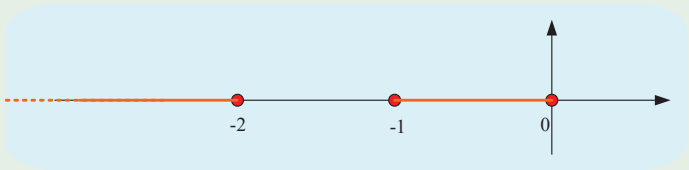


Figure 5: Available Ranges on Real Axis

Example

2. Determine the asymptotes of the root loci. When test point s is very far from the origin,

$$\lim_{s\to\infty} G(s) = \lim_{s\to\infty} \frac{K}{s(s+1)(s+2)} = \lim_{s\to\infty} \frac{K}{s^3} \tag{91}$$

and the angle condition is

$$\lim_{s\to\infty} \angle G(s) = -3\angle s = \pm\pi(2k+1) \tag{92}$$

therefore the asymptote angles could be

$$\frac{\pm\,\pi(2k+1)}{n-m} = \frac{\pm\,\pi(2k+1)}{3}, \text{ for } (k = 0, 1, 2, \cdots) \tag{93}$$

According to the mathematical condition, no matter how K varies, the asymptote angles repeat between the limited angles $\frac{\pi}{3}, \frac{-\,\pi}{3}$ and π.
When the magnitude of s goes to be very large, $G(s)$ can be also written as

$$\lim_{s\to\infty} G(s) = \lim_{s\to\infty} \frac{K}{s(s+1)(s+2)} = \lim_{s\to\infty} \frac{K}{(s+1)^3-(s+1)} = \lim_{s\to\infty} \frac{K}{(s+1)^3} \tag{94}$$

Example

therefore, $(-1, 0)$ is the intersection point of the asymptotes. So one can obtain that

$$\angle \frac{K}{(s+1)^3} = \pm\pi(2k+1) \tag{95}$$

or

$$\angle s+1 = \pm\frac{\pi}{3}(2k+1) \tag{96}$$

Substituting $s = \rho + j\omega$ into the above equation, one obtains that

$$\angle (\rho+1) + j\omega = \pm\frac{\pi}{3}(2k+1) \tag{97}$$

Taking the tangent of both sides, one obtains

$$\frac{\omega}{\rho+1} = \tan\left(\pm\frac{\pi}{3}(2k+1)\right) = \left\{\pm\sqrt{3}, \text{ and } 0\right\} \tag{98}$$

and

$$\rho + 1 = \pm\frac{\omega}{\sqrt{3}}, \text{ and } \omega = 0. \tag{99}$$

Example

Shown as in Fig.6, there are three asymptotes.

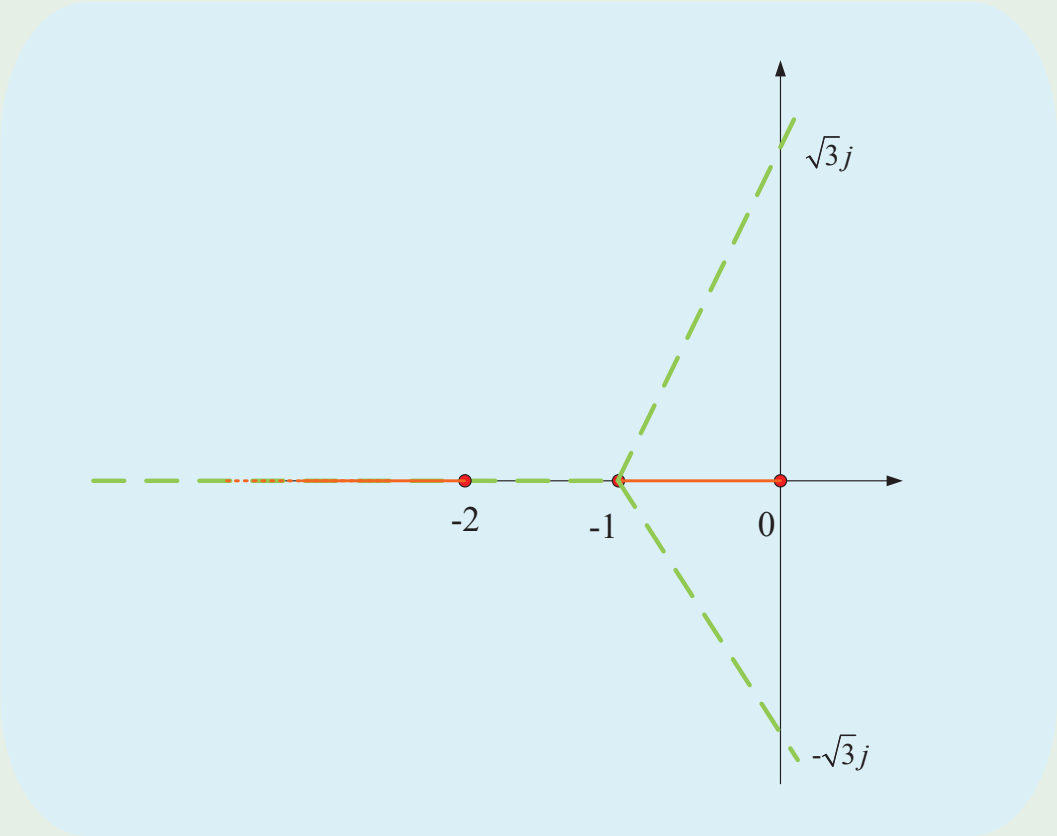


Figure 6: The Asymptotes

Example

3. Determine the breakaway point. The characteristic equation of the closed-loop system is

$$\frac{K}{s(s+1)(s+2)} + 1 = 0 \tag{100}$$

which can also be expressed as

$$s(s+1)(s+2) + K = 0 \tag{101}$$

Then, according to (83) and (84)

$$\frac{dK}{ds} = -3s^2 - 6s - 2 = 0 \tag{102}$$

and one can obtain the solution of $s = -1 \pm \frac{\sqrt{3}}{3}$. Though there are 2 solutions for (102), only $s = -1 + \frac{\sqrt{3}}{3}$ locates in one of the available ranges, $(-1, 0)$.

The corresponding value K can be calculated by substituting $s = -1 + \frac{\sqrt{3}}{3}$ back to (100) or (101).



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Example

4. Determine the points where the root loci cross the imaginary axis. The cross points on the imaginary axis are obtained by using the Routh鈥撴stability criterion. Expand the characteristic equation (101), one has that

$$s^3 + 3s^2 + 2s + K = 0 \tag{103}$$

therefore the Routh's array is

s^3	1	2
s^2	3	K
s^1	b_1	
s^0	c_1	

The parameters b_1 and c_1 are

$$b_1 = \frac{-\begin{vmatrix} 1 & 2 \\ 3 & K \end{vmatrix}}{3} = -\frac{K-6}{3}, \; c_1 = K.$$

Therefore, if $K = 6$ the crossing points on the imaginary axis can be found with the auxiliary equation in the s^2 column that

$$3s^2 + K = 3s^2 + 6 = 0 \tag{104}$$

Example

it follows that $s = \pm j\sqrt{2}$. The frequencies at the crossing points on the imaginary axis are $\omega = \pm\sqrt{2}$, and the gain value corresponding to the crossing points is $K = 6$.

Besides the above Routh's method, one can still get the crossing points on the imaginary axis by substituting $s = j\omega$ into the characteristic equation (103) that

$$(j\omega)^3 + 3(j\omega)^2 + 2j\omega + K = 0 \tag{105}$$

and another format that

$$(K - 3\omega^2) + j(2 - \omega^2)\omega = 0 \tag{106}$$

If one set the real and imaginary parts to be zero, one can obtain that

$$K - 3\omega^2 = 0, \quad (2 - \omega^2)\omega = 0 \tag{107}$$

It then follows that when $\omega = \pm\sqrt{2}, K = 6$ or $\omega = 0, K = 0$, the above equation holds. $\omega = 0, K = 0$ is on the root locus and the cross point of real and imaginary axes.



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Example

5. Shown as in Fig.7, choose a test point in the broad neighborhood of the $j\omega$ axis and the origin. Via the angle condition that $\sum_{k=1}^{3} \theta_k = \pi$, if the test points around the broad neighborhood satisfied the angle condition, the test points are on the root loci.

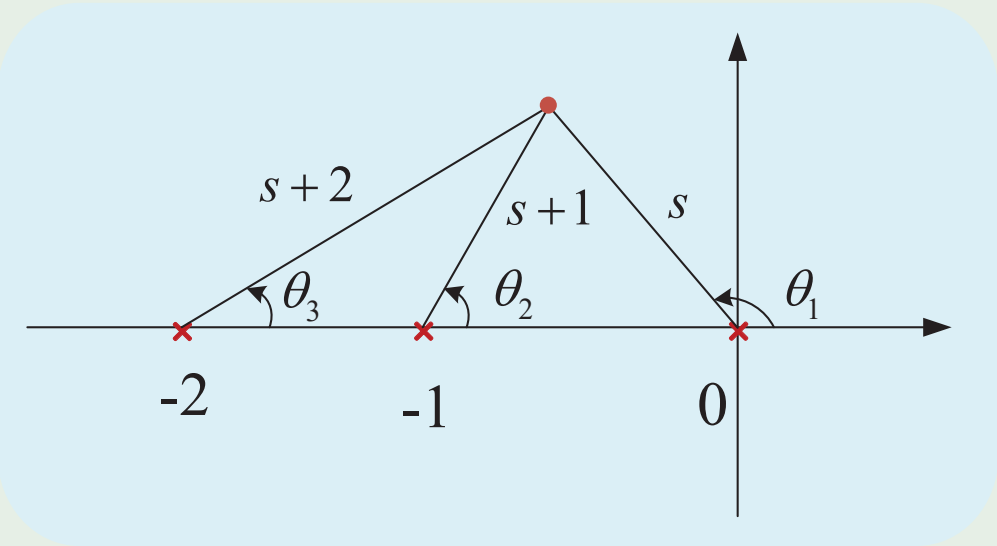


Figure 7: Angle Sum for the Test Point

Example

6. Draw the root loci. Based on the breakaway points, the asymptotes, the root locus on the real axis, and the cross point at the imaginary axis, one can further draw the root loci. The original and end points are normally marked as $'\times'$ and $'\bullet'$ respectively.

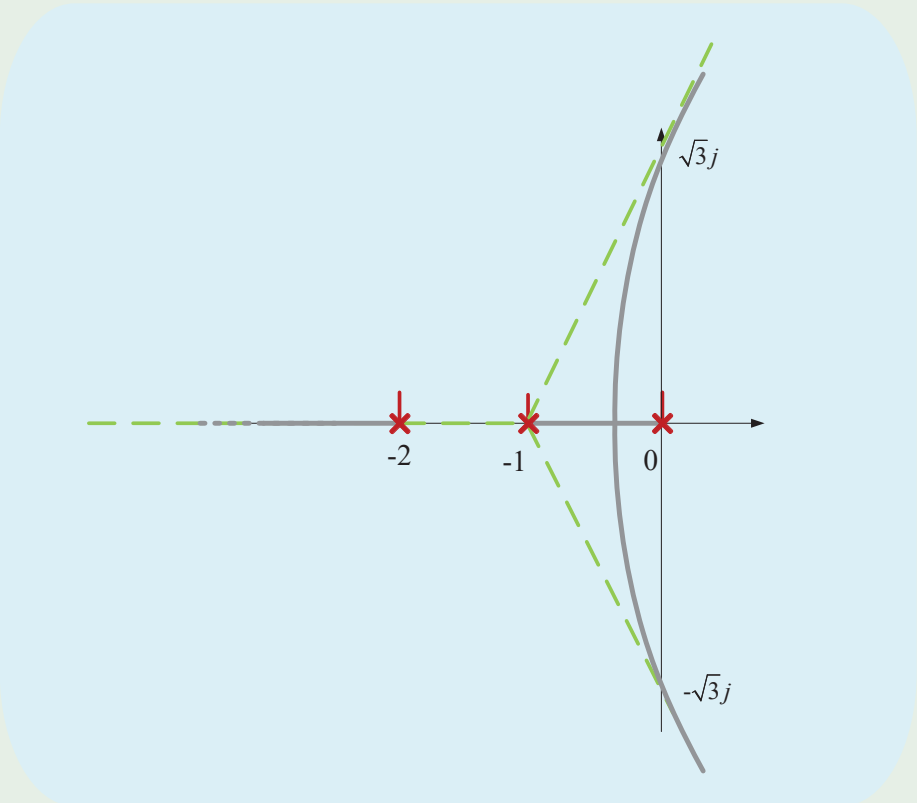


Figure 8: Root Loci Sketch

Example

7. Determine a pair of dominant complex-conjugate closed-loop poles such that the damping ratio η is 0.5.
The closed-loop poles with damping ratio $\eta = 0.5$ are located on the lines going through the origin and making the angles $\pm \arccos(\eta) = \pm \frac{\pi}{3}$ with the negative real axis, shown as in Fig. 9.

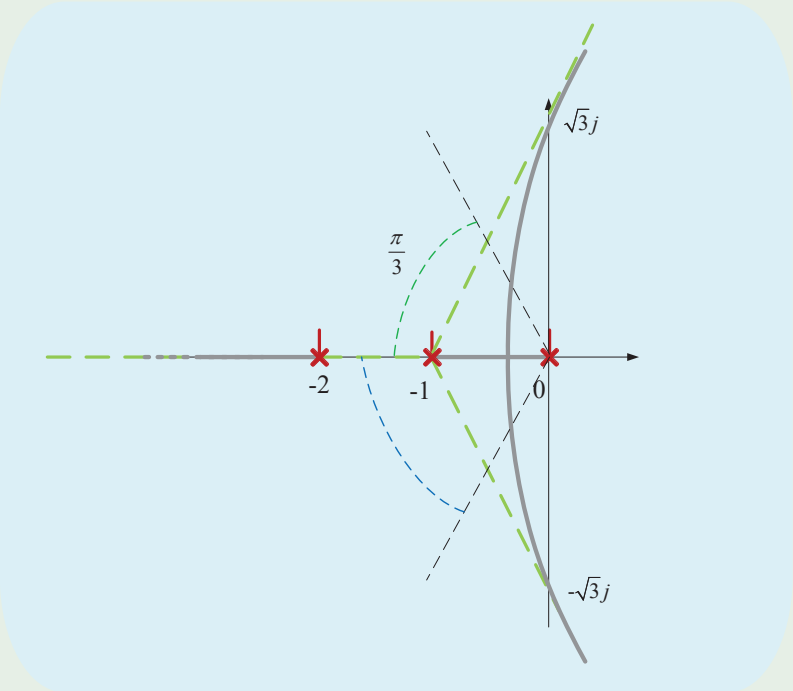


Figure 9: Poles Location based on Root Loci

Example

The accurate values of $s = -\rho \pm \sqrt{3}\rho j$, $\rho > 0$ can be calculated by substituting into the characteristic equations (100) and (101) that

$$\begin{aligned} 0 &= (-\rho + \sqrt{3}\rho j)(1 - \rho + \sqrt{3}\rho j)(2 - \rho + \sqrt{3}\rho j) + K \\ &= -6\sqrt{3}\,(\rho - 1/3)\,\rho j + 8\rho^3 - 6\rho^2 - 2\rho + K \end{aligned} \tag{108}$$

therefore, $\rho = \dfrac{1}{3}$ and $K = 1.0370$ are the solutions of above equations. Therefore, one can obtain that the cross solution for satisfying (100) are that ($s = -1/3 \pm 0.5774j$). And, the third solution in the range $(-\infty, -2]$ can be obtained by substituting K back to (100) again that $s = -2.3333$



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Example

Consider a negative feedback system in Fig.4 and $K > 0$, sketch the root locus plot

$$G(s) = \frac{K(s+2)}{s^2+2s+3}, \quad H(s) = 1. \tag{109}$$

With the given $G(s)$ and $H(s)$, one can obtain the open loop system

$$G(s)H(s) = \frac{K(s+2)}{s^2+2s+3} \tag{110}$$

which has a pair of complex-conjugate poles $s = -1 \pm \sqrt{2}j$ and a zero $s = -2$. The magnitude and phase conditions are

$$\begin{aligned} |G(s)H(s)| &= \left| \frac{K(s+2)}{s^2+2s+3} \right| = 1, \\ \underline{/G(s)} &= \underline{\left/ \frac{K(s+2)}{s^2+2s+3} \right.} = \pm\pi\,(2k+1)\,, \text{ for } k = 0, 1, 2, \cdots \end{aligned} \tag{111}$$

Example

1. The root loci on the real axis. The open-loop zero determines the location of the root locus on the real axis. Together with the phase condition and the locations of complex-conjugate poles, one can find that the available range for the root locus on the real axis is $(-\infty, -2]$.

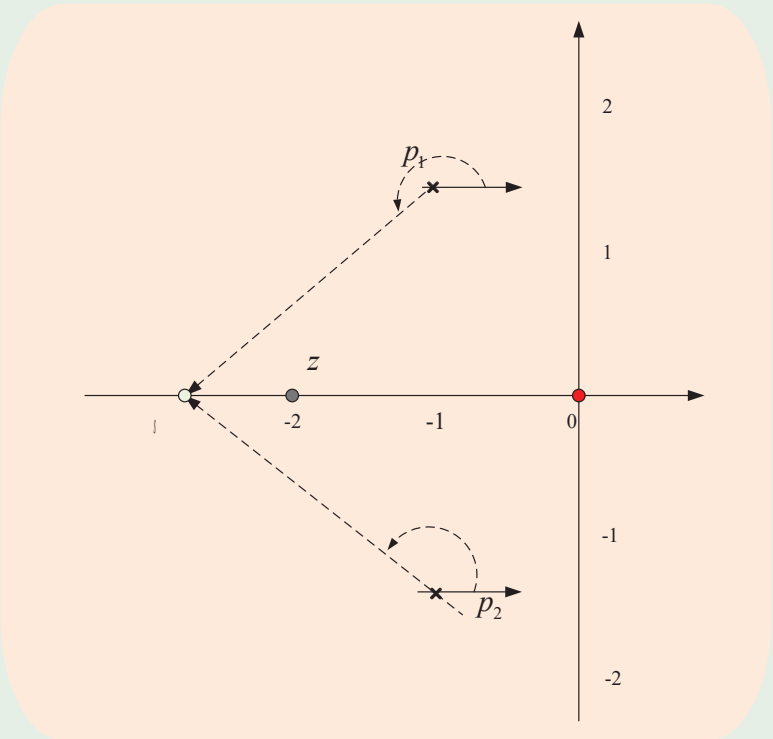


Figure 10: Test Point on the Real Axis

Example

There are two complex-conjugate poles and one zero on the negative real axis,

$$\begin{aligned} \lim_{s\to\infty} G(s)H(s) &= \lim_{s\to\infty} \frac{K(s+2)}{(s^2+2s+3)} \\ &= \lim_{s\to\infty} \frac{K}{s} \end{aligned} \tag{112}$$

whose angle condition is $\pm\pi(2k+1)$, for $(k=0,1,2,\cdots)$. And the intersect with the real axis is $(-1,0)$ by approximation that

$$G(s)H(s) = \frac{K(s+2)}{(s^2+2s+3)} = \frac{K(s+1)+K}{(s+1)^2+2} \tag{113}$$

while the magnitude of s is very large, one obtains

$$G(s)H(s) \approx \frac{K}{s+1} \tag{114}$$

one can determine that the asymptote is the negative axis.

Example

2. The departure angle from the complex-conjugate open-loop poles.
The sum angles should satisfy

$$\sum_{i=1}^{m} \phi_i - \sum_{j=1}^{n} \theta_j = \pi(2k+1), \ (k = 0, 1, 2, 3, \cdots) \tag{115}$$

one has that

$$\phi_1 - (\theta_1 + \theta_2) = \pi(2k+1) \tag{116}$$

For the angle of test point near the pole p_1, the angle $\phi_1 = \arctan\sqrt{2}$, one has that

$$\arctan\sqrt{2} - \theta_1 - \frac{\pi}{2} = \pm\pi(2k+1) \tag{117}$$

and one can verify that

$$\theta_1 = \pm 2k\pi + \frac{\pi}{2} + \arctan\sqrt{2} \approx 144.735^\circ. \tag{118}$$

Since the root locus is symmetric about the real axis, the angle of departure from the pole at p_2 is about -144.735°.

Example

3. The break-in point. For the break-in point calculation, with the characteristic function one can obtain the functional of $K(s)$ that

$$K = -\frac{s^2 + 2s + 3}{s + 2} \tag{119}$$

one has the differential of K that

$$\frac{dK}{ds} = -\frac{(2s + 2)(s + 2) - (s^2 + 2s + 3)}{(s + 2)^2} = -\frac{s^2 + 4s + 1}{(s + 2)^2} \tag{120}$$

The solution for the equation $s^2 + 4s + 1 = 0$ are $s = -0.2679, s = -3.7321$, but only $s = -3.7321$ is at the available range $(-\infty, -2]$. And the corresponding gain value is $K = 5.4641$



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Example

4. Sketch the root-locus plot. With the above information, one can sketch the root locus as in following Fig.11.

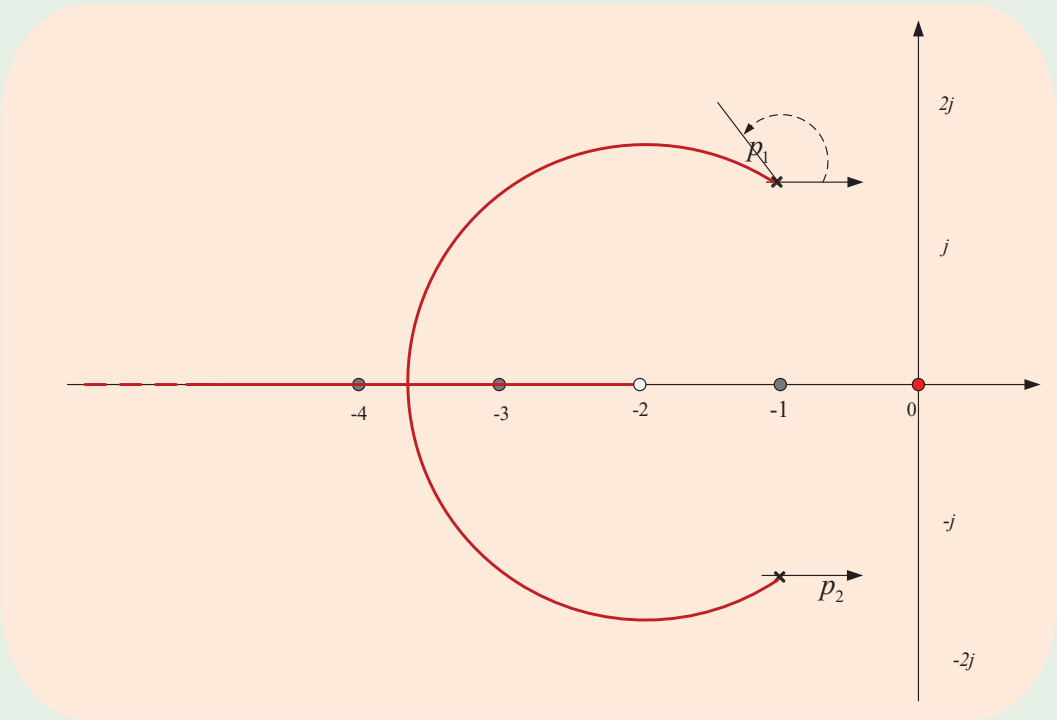


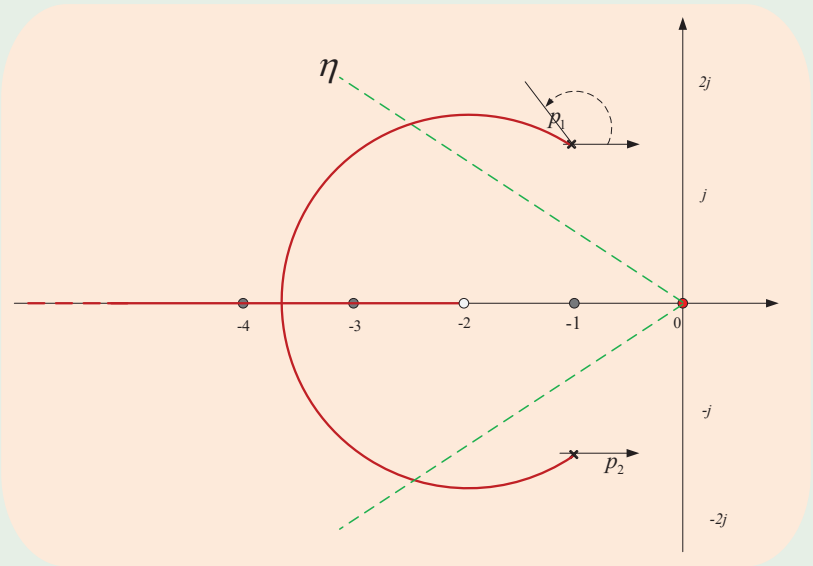
Figure 11: Root Loci Plot

Example

Similar as in former Example, one can further calculate the root loci with different damping ratios η as in Fig.12, with the conjugate pair $s = -\rho \pm \frac{\sqrt{1-\eta^2}}{\eta}\rho j$, $\rho > 0$, by substituting $s = -\rho + \frac{\sqrt{1-\eta^2}}{\eta}\rho j$ or $s = -\rho - \frac{\sqrt{1-\eta^2}}{\eta}\rho j$ back to the characteristic equation

$$K(s+2) + (s^2 + 2s + 3) = 0 \tag{121}$$

whose real and imaginary parts are both zeros.



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✠Different with the negative one, **the transfer function of positive feedback system** is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)} \tag{122}$$

whose characteristic equation is

$$1 - G(s)H(s) = 0, \text{ or } G(s)H(s) = 1 \tag{123}$$

✠**The magnitude and phase conditions** are

$$|G(s)H(s)| = 1, \text{ and } \angle G(s)H(s) = 2k\pi, \quad \text{for } (k = 0, 1, 2, \cdots). \tag{124}$$

✠**The angles of asymptotes** are

$$\frac{\pm 2k\pi}{n - m}, \text{for } k = 0, 1, 2, \cdots \tag{125}$$

where n and m indicate the numbers of finite poles and zeros of open loop system $G(s)H(s)$.

✠**The departure angles** near poles and zeros are calculated based on the phase sum $\pm 2k\pi$.

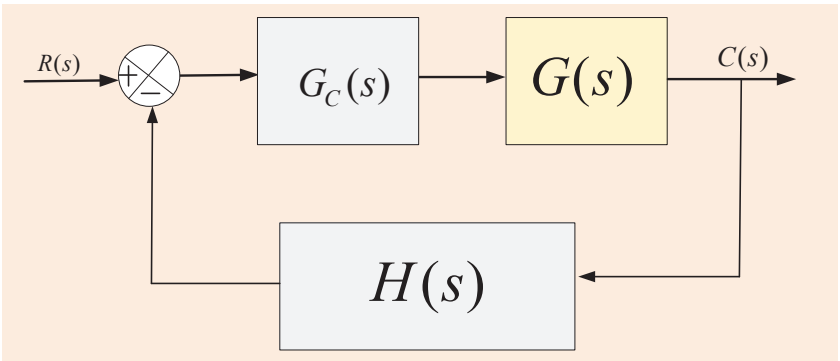
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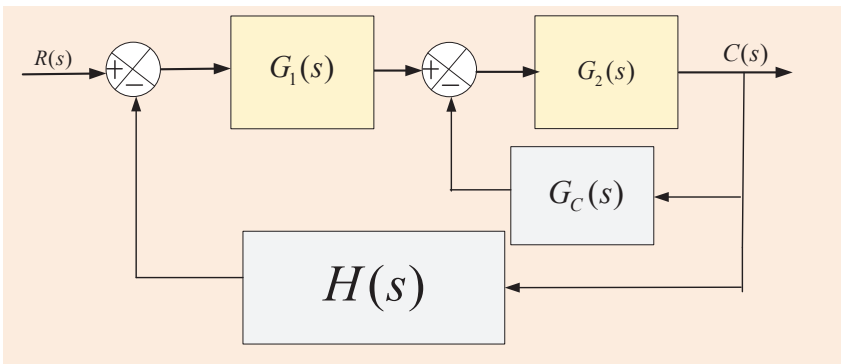
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Typical Root Locus based Design Methods

- Poles and Zeros Addition;
- Lead-Lag Compensation;
- Gain Design;
- Serial and Parallel Compensation.



(a) Serial Compensator



(b) Parallel Compensator

Figure 13: Common Compensators

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Example

For an open-loop system as in Fig.13(a), whose transfer function is

$$G(s) = \frac{1}{s(s+1)(s+2)} \tag{126}$$

if $H(s) = 1$, consider the serial compensators $G_c(s)$ whose transfer function are in following forms respectively

$$G_{C1}(s) = K,\ G_{C2}(s) = K(s+3),\ \text{and}\ G_{C3}(s) = K\frac{1}{s+3},\ \text{for}\ K > 0. \tag{127}$$

1. Compare the root locus variations of compensated systems under different compensators.
2. Set $K = 1$, compare the step responses, bode diagrams, and Nyquist plots of the closed-loop systems under unit negative feedback.



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Example

The root locus plots of different open loop systems $G_1(s) = G(s)G_{C1}(s)$, $G_2(s) = G(s)G_{C2}(s)$ and $G_3(s) = G(s)G_{C3}(s)$ for different gain K are depicted as in Fig.14. The compensation results under different pole and zero addition show that, the pole addition method can pull the root locus to the right. The zero addition can pull the root locus to the left.

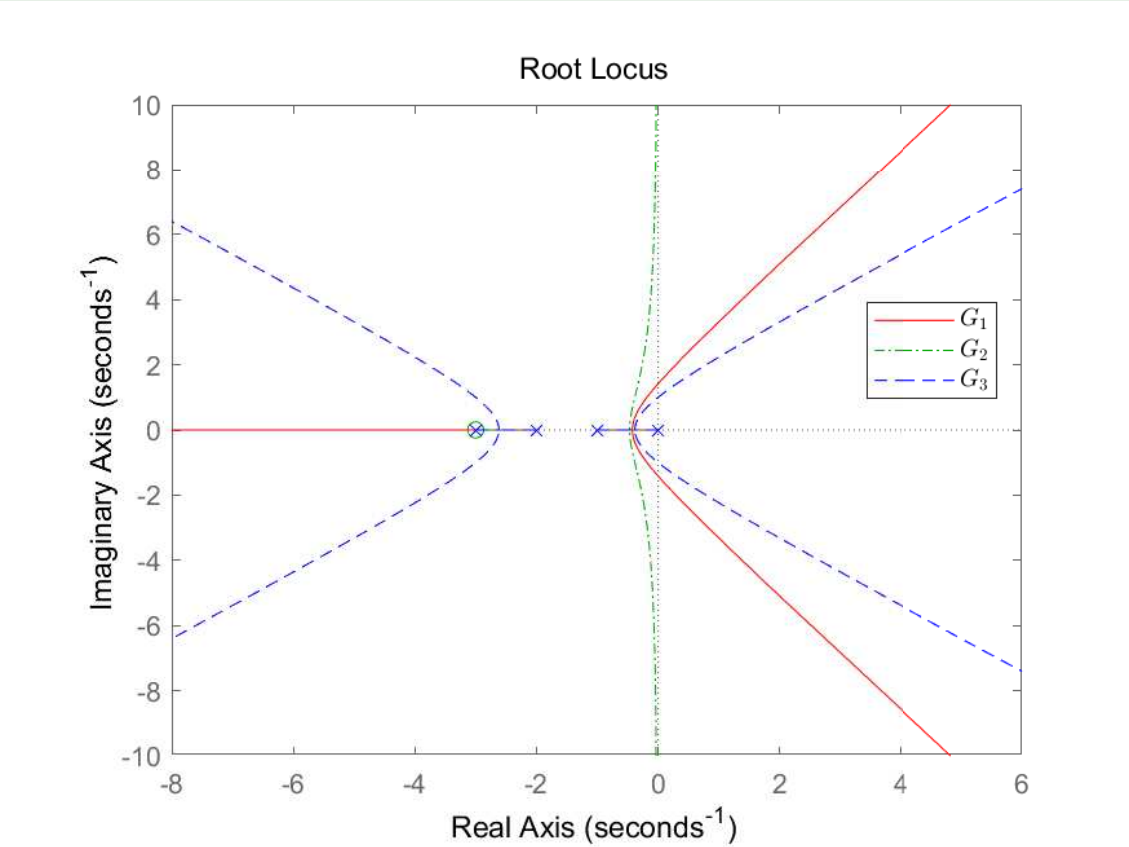
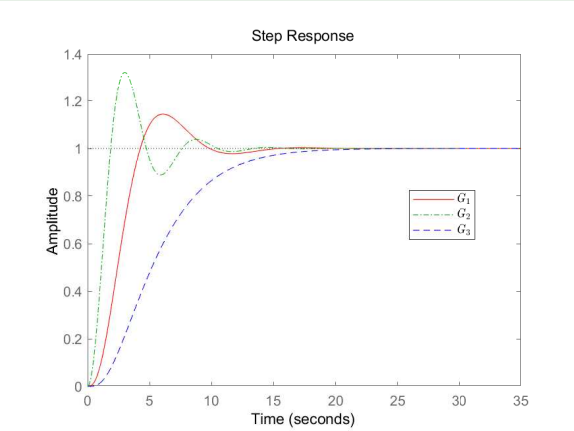


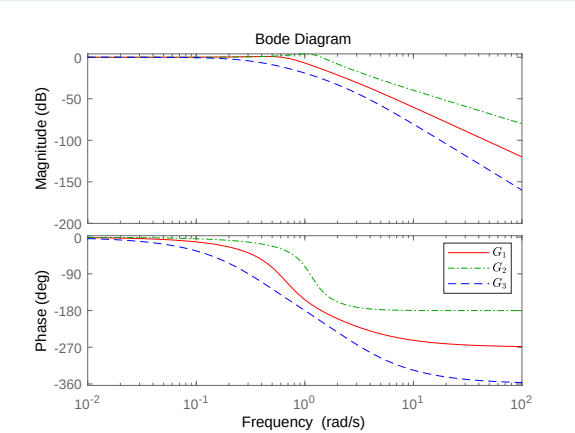
Figure 14: Root Locus Plots under Different Compensators

Example

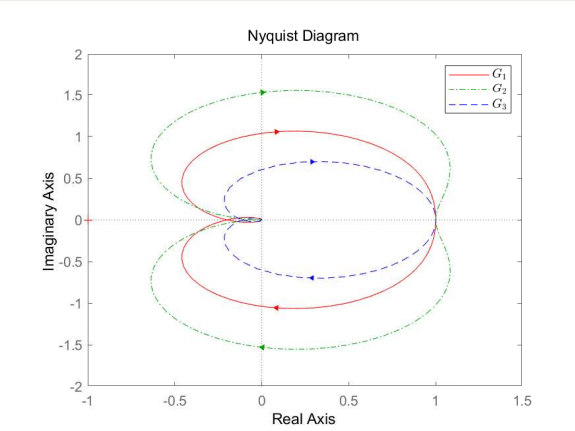
Adding a zero speeds up the transient response, adds phase lead, and increases the system magnitude. Adding pole to the system can slow down the transient response, lower the phase and magnitude.



(a) Step Response of Pole and Zero Compensation



(b) Bode Diagram of Pole and Zero Compensation



(c) Nyquist Plot of Pole and Zero Compensation

Figure 15: Pole and Zero Compensation of Closed-loop System

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Transfer function with an amplifier electrical circuit based compensator in Fig.16 is

$$\frac{U_3(s)}{U_1(s)} = K_c \frac{s + \dfrac{1}{T}}{s + \dfrac{1}{\alpha T}} \tag{128}$$

where

$$K_c = \frac{R_4 C_1}{R_3 C_2},\ T = R_1 C_1,\ \text{and } \alpha = \frac{R_2 C_2}{R_1 C_1}$$

When $\alpha < 1$ or $\alpha > 1$ (128) would be lead and lag systems respectively.

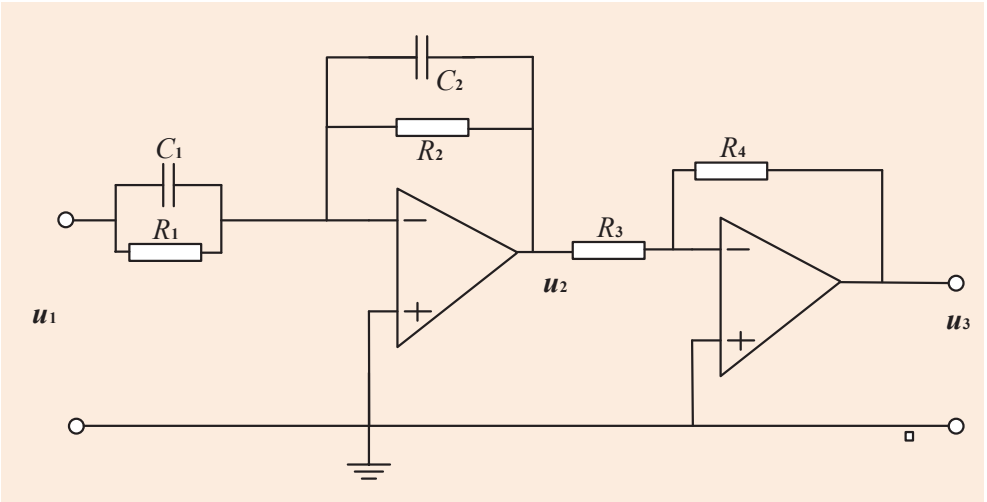


Figure 16: System Block with Lag and Lead Compensation

Example

An open-loop system in Fig.13(a), whose transfer function is

$$G(s) = \frac{1}{s(s+1)(s+2)} \tag{129}$$

if $H(s) = 1$, consider the serial compensators $G_c(s)$ in following forms

$$G_{C1}(s) = K\frac{s+\frac{1}{T}}{s+\frac{1}{\alpha_1 T}}, \text{ and } G_{C2}(s) = K\frac{s+\frac{1}{T}}{s+\frac{1}{\alpha_2 T}} \tag{130}$$

where $T = 0.1$, $\alpha_1 = 0.5$ and $\alpha_2 = 2.5$.

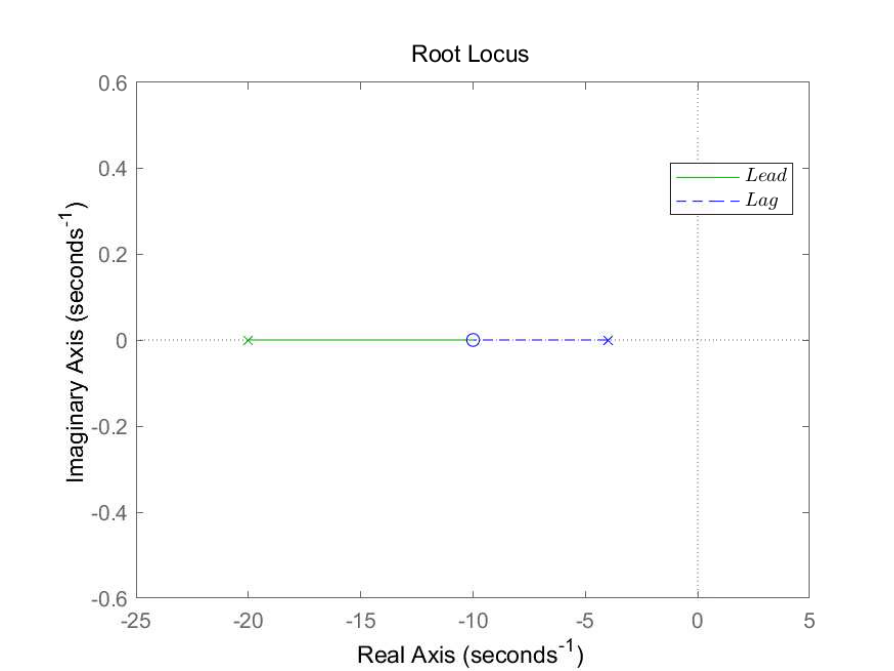
1. Compare the root locus variations of compensated systems under lead and lag compensators.
2. Set $K = 1$, compare the step responses, bode diagrams, and Nyquist plots of the closed-loop systems under unit negative feedback.



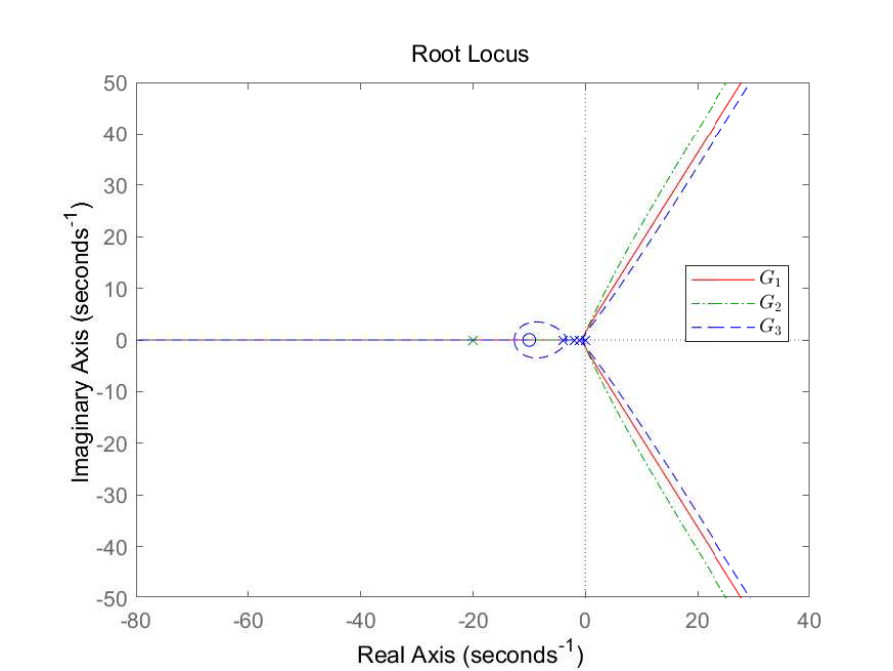
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Example

The root locus of the lead block G_{C1} and lag blockG_{C2} are compared in Fig.17(a).
The root locus of $G(s)H(s)$ combined with the lead and lag blocks are in Fig.17(b).



(a) Root Locus of Pure Lead and Lag Compensation

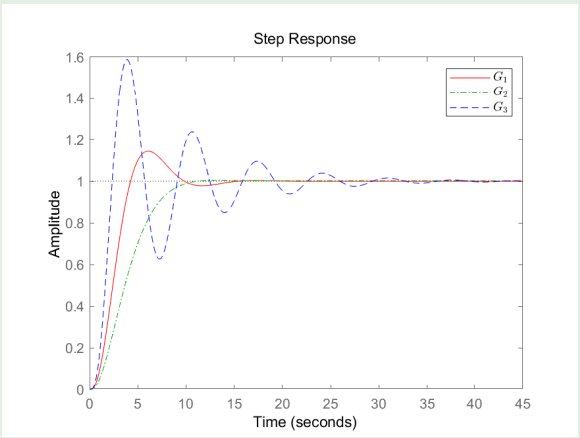


(b) Root Locus of Open Loop System with Lead and Lag Compensation

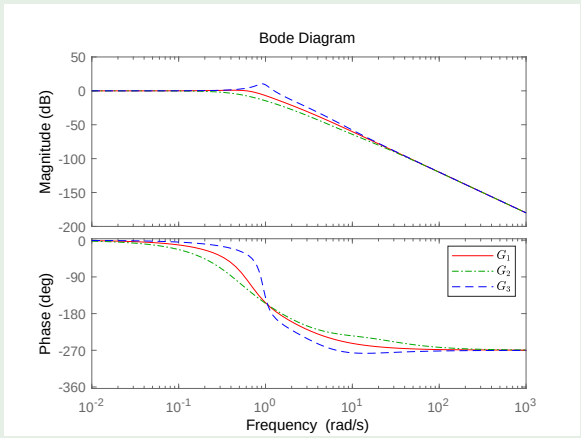
Figure 17: Lead and Lag Compensation of Open System

Example

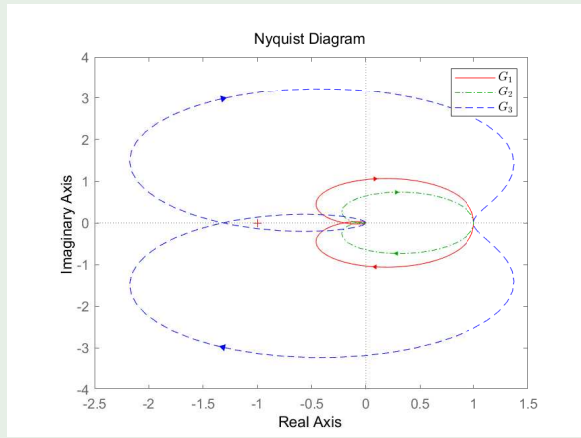
The step responses, the bode diagram, and the Nyquist plots, for the original, lead combined, and lag combined closed-loop system, are



(a) Step Responses



(b) Bode Diagrams



(c) Nyquist Plots

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7 Lyapunov Stability Criteria

- Concept Definition
- Lyapunov's Second Method

Consider an autonomous nonlinear dynamical system

$$\dot{x} = f(x(t)), \qquad x(0) = x_0 \tag{131}$$

where $x(t) \in \mathcal{D} \subseteq \Re^n$ denotes the system state vector, $\mathcal{D}$ is an open set containing the origin, and $f : \mathcal{D} \to \Re^n$ is a continuous vector field on $\mathcal{D}$. Suppose f has an equilibrium at x_e so that $f(x_e) = 0$ then

1. This equilibrium is said to be **Lyapunov stable**, if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that, if $\|x(0) - x_e\| < \delta$, then for every $t \geq 0$ we have $\|x(t) - x_e\| < \epsilon$.
2. The equilibrium of the above system is said to be **asymptotically stable** if it is Lyapunov stable and there exists $\delta > 0$ such that if $\|x(0) - x_e\| < \delta$, then $\lim_{t\to\infty} \|x(t) - x_e\| = 0$.
3. The equilibrium of the above system is said to be **exponentially stable** if it is asymptotically stable and there exist $\alpha > 0, \beta > 0, \delta > 0$ such that if $\|x(0) - x_e\| < \delta$, then $\|x(t) - x_e\| < \alpha\|x(0) - x_e\|e^{-\beta t}$ for all $t \geq 0$.



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7 Lyapunov Stability Criteria

- Concept Definition
- Lyapunov's Second Method

Lyapunov Function

Consider a radial unbounded function $V : \mathbb{R}^n \to \mathbb{R}$ such that

- $V(x) = 0$, if and only if $x = 0$
- $V(x) > 0$, if and only if $x \neq 0$
- $\dot{V}(x) = \dfrac{dV(x)}{dt} = \sum_{i=1}^{n} \dfrac{\partial V(x)}{\partial x_i} f_i(x) = \bigtriangledown V \bullet f(x) \leq 0$

Then $V(x)$ is a Lyapunov function and the system is stable in the Lyapunov sense.

- Condition $V(0) = 0$ is required, otherwise function like $V(x) = \dfrac{1}{1 + |x|}$ can prove that $\dot{x} = x$ is locally stable.
- Additional condition "properness" or "radial unboundedness", i.e. when $x \to \infty$, $V(x) \to \infty$, is required for global stability and global asymptotic stability.



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Example

Examine the stability of following systems

$$\dot{x} = -x \tag{132}$$

Construct a radial unbounded Lyapunov function $V(x) = x^2(t)$, i.e.$\lim_{x\to\infty} V(x) = \infty$. And it satisfies that

$$V(0) = 0, \text{and } V(x) > 0 \text{ when } x \neq 0 \tag{133}$$

it can further verify that

$$\begin{aligned} \dot{V}(x) &= \frac{dV(x)}{dt} = \frac{\partial V(x)}{\partial x}\frac{d\left(x\left(t\right)\right)}{dt} \\ &= 2x\dot{x}(t) \\ &= -2x^2(t) \\ &< 0, \text{when } x \neq 0. \end{aligned} \tag{134}$$

One can determine that (132) is stable in Lyapunov sense. And, (132) is also global asymptotically stable.

Example

Consider the following dynamic system

$$\begin{cases} \dot{x}_1 &= -x_1 + 4x_2 \\ \dot{x}_2 &= -x_1 - x_2^3 \end{cases} \tag{135}$$

The only equilibrium point for this system is the origin $x = 0$. Set a quadratic Lyapunov function

$$V = x_1^2 + \alpha x_2^2 \tag{136}$$

where α is a positive constant parameter can be determined alternatively. As in former Example.64, one determines that $V(0) = 0$, and $\lim_{x\to\infty} V(x) = \infty$. One can further determine that

$$\dot{V} = \begin{bmatrix} 2x_1 & 2\alpha x_2 \end{bmatrix} \begin{bmatrix} -x_1 + 4x_2 \\ -x_1 - x_2^3 \end{bmatrix} \tag{137}$$

And that

$$\dot{V} = -2x_1^2 + (8 - 2\alpha)x_1 x_2 - 2\alpha x_2^4 \tag{138}$$

therefore, if one selects $\alpha = 4$, then $\dot{V} = -2x_1^2 - 2\alpha x_2^4$. When $x \neq 0$, one can determine that $\dot{V} < 0$ holds, and (135) is asymptotically stable. And one can determine that $x = 0$ is also a globally asymptotically stable equilibrium point.

Example

Consider a nonlinear system below, whose equations are described as

$$\begin{cases} \dot{x} &= \rho(y-x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz \end{cases} \tag{139}$$

where ρ, r and b are positive constants. Analyze the stability of the origin under the condition $r < 1$. Define the Lyapunov function below

$$V = \alpha_1 x^2 + \alpha_2 y^2 + \alpha_3 z^2 \tag{140}$$

α_1, α_2 and α_3 are positive parameters to be determined. One can find that $V > 0$, $V = 0$, when $x = 0$, and V is radially unbounded, i.e. $\lim_{x\to\infty} V(x) = \infty$. The derivative of V along the trajectories of the system is

$$\begin{aligned} \dot{V} &= \begin{bmatrix} 2\alpha_1 x & 2\alpha_2 y & 2\alpha_3 z \end{bmatrix} \begin{bmatrix} \rho(y-x) \\ rx - y - xz \\ xy - bz \end{bmatrix} \\ &= 2xy(\alpha_1\rho + \alpha_2 r) - 2\alpha_1\rho x^2 - 2\alpha_2 y^2 - 2\alpha_3 b z^2 - 2xyz(\alpha_2 - \alpha_3) \end{aligned} \tag{141}$$

Example

Then, if one selects $\alpha_1 = \frac{1}{\rho}$, $\alpha_2 = 1$ and $\alpha_3 = 1$, one verifies that

$$\begin{aligned} \dot{V} &= 2xy(1+r) - 2x^2 - 2y^2 - 2bz^2 \\ &= -2\left(x^2 + y^2 + bz^2 - (1+r)xy\right) \\ &= -2\left[\left(x - \frac{1}{2}(1+r)y\right)^2 + \left(1 - \left(\frac{1+r}{2}\right)^2\right)y^2 + bz^2\right] \end{aligned} \tag{142}$$

With the parameter $0 < r < 1$, one can determine that $\dot{V} < 0$ holds. And the system is asymptotically stable if the Lyapunov parameters $\alpha_1 = \frac{1}{\rho}$, $\alpha_2 = 1$ and $\alpha_3 = 1$.



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Example

Consider a pendulum of mass M at the end of a rigid massless rod of length R:

$$MR\ddot{\theta} + Mg\sin(\theta) = 0 \tag{143}$$

where θ is the angle made with the downward direction, and g is the acceleration due to gravity. Let $x_1 = \theta$, and $x_2 = \dot{\theta}$, then

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\dfrac{g}{R}\sin(x_1) \end{cases} \tag{144}$$

Here, one can define the Lyapunov function that

$$V(x) = \frac{1}{2}MR^2x_2^2 + MgR\left(1 - \cos(x_1)\right) \tag{145}$$

It satisfies the radial unbounded condition, $\lim_{x\to\infty} V(x) = \infty$, $V(x = 0) = 0$ and $V(x) > 0, \forall x \neq 0$. Then, for the derivative of $V(x)$, one obtains that

$$\begin{aligned} \dot{V}(x) &= MR^2x_2\dot{x}_2 + MgR\sin(x_1)\dot{x}_1 \\ &= -MR^2\frac{g}{R}x_2\sin(x_1) + MgR\sin(x_1)x_2 \\ &= 0 \end{aligned} \tag{146}$$

Example

One cannot determine the stability of system (144) with the Lyapunov function (145) given in this example.

Beside the system considered in (143), the damping torque which is proportional to the velocity is added to the pendulum system, the state-space model would be

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -Dx_2 - \dfrac{g}{R}\sin(x_1) \end{cases} \tag{147}$$

With the same Lyapunov function as in (145), one can obtain the derivative of $V(x)$ that

$$\dot{V} = -DMR^2x_2^2 \leq 0 \tag{148}$$

One can determine that (147) is asymptotically stable under the same Lyapunov function.



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5 Nyquist Stability Criteria

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7 Lyapunov Stability Criteria

8 Robust Stability

9 Chapter Conclusion

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Why Robust Stability?

Robust stability indicates that the closed-loop control system should not reach an unstable behavior even in the worst case of the parameter changes.

The controller of the closed-loop system is required to be stable through the given parameter uncertainty ranges. The mismatch between the model and the system is *model uncertainty*.

Consider the simplified satellite below

$$I\ddot{\theta} = Fd + \tau_d \tag{149}$$

whose dynamics is

$$(I_n + \Delta_I)\ddot{\theta} = Fd \tag{150}$$

where the model uncertainty $\Delta_I \in \left[-\overline{\Delta}_I, \overline{\Delta}_I\right]$, $\overline{\Delta}_I > 0$. The robust stability analysis for the

system (150) is that the system under controller $F = F_c$, for any uncertain value $\Delta_I \in \left[-\overline{\Delta}_I, \overline{\Delta}_I\right]$, is still stable. If one selects the controller $F = \frac{-k_1\dot{\theta} - k_2\theta}{d}$, the closed-loop system considering the model uncertainties can be

$$(I_n + \Delta_I)\ddot{\theta} + k_1\dot{\theta} + k_2\theta - \tau_d = 0 \tag{151}$$



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Example

If the parameters in (151) are $I_n = 1\ \text{kg}\cdot\text{m}^2$, $\overline{\Delta}_I = 0.2\cdot I_n$, $k_1 = 5$, $k_2 = 10$, $d = 1$, analyze the system stability.

$$(1+\Delta_I)\ddot{\theta} + 5\dot{\theta} + 10\theta = 0 \tag{152}$$

The initial states are $\theta(0) = 0$, $\dot{\theta}(0) = 0$, and $\ddot{\theta}(0) = 0$. Then the transfer function model of (152) would be

$$G(s) = \frac{1}{(1+\Delta_I)s^2 + 5s + 10} \tag{153}$$

According to the characteristic equation, one obtains the Routh array

s^2	$1+\Delta_I$	10
s^1	5	

The sufficient condition to guarantee the system stability of (152), is $1+\Delta_I > 0$, i.e. $\Delta_I > -1$.

Though the system parameter $I = I_n + \Delta_I$ is uncertain, for the given uncertain range $\Delta_I \in (-0.2I_n, 0.2I_n)$, the controller still stabilizes the system.



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- Asymptotic Stability; Lyapunov Stability; Poles located in the left half plane;
- Bode Diagram(Open and Closed), Polar Plot (ω varies), and Root Locus (Poles location when K changes);
- Controller: Lead and Lag Compensator;

Thanks!

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